## STRUCTURAL DYNAMICS

**2.0 Single Degree of Freedom (SDOF) System (20 hours) Contd.**

2.1 Equations of Motion: Modeling of single-degree-of-freedom system, response parameters, effect of gravity, effect of support excitation, combination of stiffness (Parallel and series)

2.2 Free vibration response: Undamped system; Damped system (Under critically, critically and over critically)

2.3 Logarithmic decrement of response

2.4 Forced vibration response to Harmonic Loading: Undamped system; damped system (Transient response, steady-state response, general response and resonance response)

2.5 Energy Dissipated by Damping (viscous damping)

2.6 Forced Vibration Response to Periodic Loading: Undamped system (Transformation of loading and response using Fourier series)

2.7 Forced Vibration Response to Impulsive load (half-sine wave pulse, rectangular pulse, ramp pulse, triangular pulse)



## STRUCTURAL DYNAMICS

**2.0 Single Degree of Freedom (SDOF) System (20 hours)**

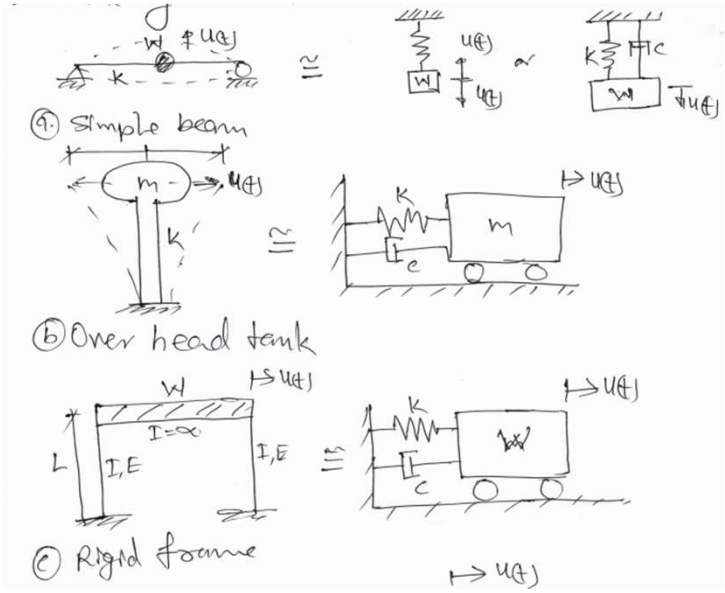
2.8 Forced Vibration Response to General Dynamic Loading: Unit-impluse and Dirac-delta function, Unit-impulse response function, Duhamel integral and convolution integral, Numerical evaluation of Duhamel integral

2.9 Vibration measuring instruments (Displacement meter and accelerometer)

2.10 Vibration isolation (Vertical oscillating force, harmonic motion of the base, force transmissibility, unbalanced mass amplitude of rotating unbalanced system)

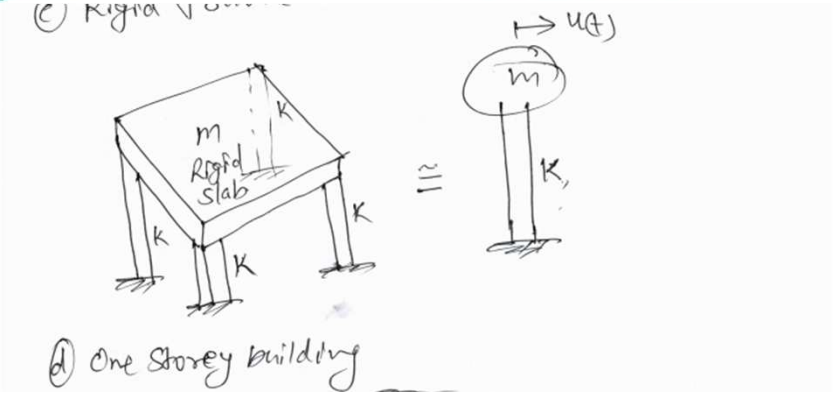
2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

## 2.1 Equations of Motion: Modeling of single-degree-of-freedom system






## 2.1 Equations of Motion: Modeling of single-degree-of-freedom system



## 2.1 Equations of Motion: Response parameters

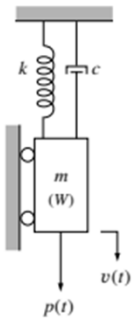
These are the important response parameters of SDOF systems:

1. **Natural Frequency ($\omega_n$, $f_n$)**
   The frequency at which the system vibrates freely after being disturbed.
2. **Natural Period (Tn)**
   Time required to complete one cycle of free vibration'
3. **Damping Ratio ($\xi$)**
   Indicate the amount of damping relative to critical damping
4. **Damped Natural Frequency ($\omega_d$, $f_d$)**
   Frequency of vibration having damping.
5. **Maximum Displacement ($d_{max}$)**
   Peak displacement under dynamic loading
6. **Dynamic amplification factor ($\rho$)**
   Ratio of dynamic response to static response
7. **Resonance**
   Occurs when forcing frequency approaches natural frequency
8. **Logarithmic Decrement ($\delta$)**
   Used to evaluate damping from free vibration measurements.

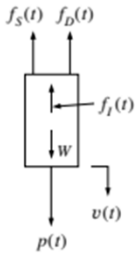


## 2.1 Equations of Motion: Effect of gravity

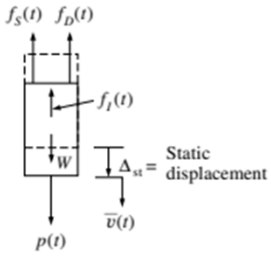
Consider a system as shown so that the force of gravity acts in the direction of the displacement.




SDOF System




Free Body Diagram




Free Body diagram with static displacement

## 2.1 Equations of Motion: Effect of gravity

Using equilibrium equation:

$$m\,\ddot{v}(t) + c\,\dot{v}(t) + k\,v(t) = p(t) + W \quad \ldots\ldots 1$$

But total displacement, u(t) = $\Delta$st + $\bar{u}$(t) $\ldots\ldots 2$

And spring force, fs(t) = ku(t) = k$\Delta$st + k$\bar{u}$(t) $\ldots\ldots 3$

Substituting value of u(t) in equation (1),

m.$\ddot{u}$(t) + c.$\dot{u}$(t) + k$\Delta$st + k$\bar{u}$(t) = p(t) + W

Noting that, k$\Delta$st $\cong$ W, then

m.$\ddot{u}$(t) + c.$\dot{u}$(t) + k$\bar{u}$(t) = p(t) $\ldots\ldots (4)$



## 2.1 Equations of Motion: Effect of gravity

Now differentiating equation (2), Noting that $\Delta$st does not vary with time, it is evident that,

$$\ddot{v}(t) = \ddot{\bar{v}}(t) \text{ and } \dot{v}(t) = \dot{\bar{v}}(t)$$

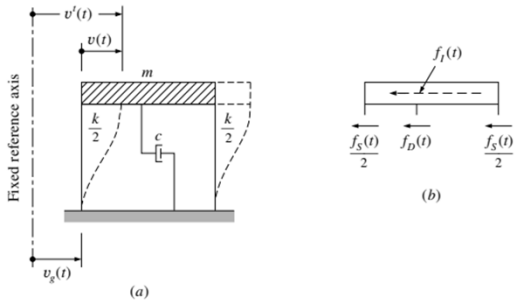
So that equation (4) can be written as

$$m\,\ddot{\bar{v}}(t) + c\,\dot{\bar{v}}(t) + k\,\bar{v}(t) = p(t) \quad \ldots (5)$$

Comparing equation (5) and equation of motion of SDOF system with p(t) demonstrates that the equation of motion expressed with reference to static equilibrium position of the dynamic system is not affected by gravity force.

## 2.1 Equations of Motion: Effect of support excitation

A simplified model of the earthquake excitation problem is shown in figure below, in which the horizontal ground motion caused by the earthquake is indicated by the displacement $u_g(t)$ of the structure's base relative to the fixed reference.




Horizontal girder of the frame is assumed to be rigid and to include all the moving mass of the structure. The vertical columns are assumed to be weightless and inextensible in the vertical direction.



## 2.1 Equations of Motion: Effect of support excitation

The equilibrium of forces for this system can be written as

$$f_I(t) + f_D(t) + f_s(t) = 0 \quad \ldots \quad \ldots (1)$$

In which, damping force, $f_D(t) = c\dot{u}(t)$

elastic force, $f_s(t) = ku(t)$,

However, the inertial force in this case is given by

$$f_I(t) = m\ddot{u}^t(t)$$

Where $u^t(t)$ represent the total displacement of the mass from the fixed reference axis. Substituting for the inertial, damping and elastic force in above equation no (1) yields

$$m\ddot{u}^t(t) + c\dot{u}(t) + ku(t) = 0 \ldots \quad \ldots (2)$$

But, $u^t(t) = u(t) + u_g(t) \quad \ldots \quad \ldots (3)$,

And $\dot{u}^t(t) = \dot{u}(t) + \dot{u}_g(t)$,

And $\ddot{u}^t(t) = \ddot{u}(t) + \ddot{u}_g(t)$

## 2.1 Equations of Motion: Effect of support excitation

Substituting in equation no (2), yields

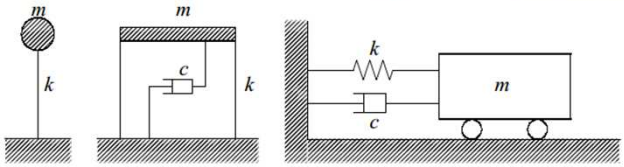
$m\ddot{u}(t) + m\ddot{u}_g(t) + c\dot{u}(t) + ku(t) = 0$

OR $m\ddot{u}(t) + c\dot{u}(t) + ku(t) = -m\ddot{u}_g(t) \cong p_{eff}(t)$ … (4)

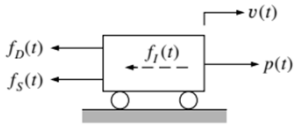
Where, $p_{eff}(t)$ denotes the effective support excitation loading. –ve sign denotes that effective force opposes the sense of ground acceleration.



## 2.1 Equations of Motion




Consider SDOF system as shown above.
Free body diagram of the system,




Equilibrium of mass in motion, $\sum F = 0$

$f_s(t) + f_D(t) + f_I(t) = f(t)$ … … (1), or

$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = f(t)$ … … (2)

The above equation (2) is called equation of motion for SDOF system.

## 2.1 Equation of Motion

Where, $f_s(t)$ = Spring force = $ku(t)$
$f_D(t)$ = Damping force = $c\dot{u}(t)$
$f_I(t)$ = Inertia force = $m\ddot{u}(t)$

If applied force, $f(t) = 0$, then
$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = 0$ … … (3)
Free vibration equation of motion for SDOF system.

If damping coeficient, $c = 0$, then
$m\ddot{u}(t) + ku(t) = 0$ … … (4)
Undamped free vibration equation of motion for SDOF system.

Equation (4) is ordinary differential equation.
Equation (2) is linear differential equation of second degree with constant coefficient.
If $f(t) = 0$, the equation become homogeneous
If $f(t) \neq 0$, it is non-homogeneous



## 2.2 Free vibration response: Undamped system

Undamped free vibration:

$m\ddot{u} + ku = 0$

Assuming a trial solution

$u = A\cos\omega t$ (or $u = B\sin\omega t$)

$\therefore \dot{u} = \omega A \sin\omega t$

& $\ddot{u} = -\omega^2 A\cos\omega t$

$\therefore m(-\omega^2 A\cos\omega t) + kA\cos\omega t = 0$

or $(-m\omega^2 + k)A\cos\omega t = 0$

But $A\cos\omega t \neq 0$

## 2.2 Free vibration response: Undamped system

$\therefore -m\omega^2 + k = 0$

$\omega^2 = \frac{k}{m}$

and positive root of this eqn.

$\omega = \sqrt{\frac{k}{m}}$ is known as natural frequency.

Also, $W = mg$

$\therefore \omega = \sqrt{\frac{kg}{W}}$

If $u_{st} = W/k$, static displacement due to weight $W$.

then, $\omega = \sqrt{\frac{g}{u_{st}}}$



## 2.2 Free vibration response: Undamped system

$m\ddot{u}(t) + k\,u(t) = 0$ —①

Solution of this second order differential equation with coefficient (constant) has the form

$u(t) = e^{st}$ , $s$ = unknown constant

$\therefore \dot{u}(t) = s\,e^{st}$

and, $\ddot{u}(t) = s^2 e^{st}$

Substituting in equation ①

$m(s^2 e^{st}) + k(e^{st}) = 0$

or, $(ms^2 + k)e^{st} = 0$

$\because e^{st} \neq 0$, $\therefore ms^2 k = 0$ or, $s^2 = -\frac{k}{m} = i^2\frac{k}{m}$

and: $\boxed{s = \pm i\omega}$ or $s_1 = i\omega$ & $s_2 = -i\omega$

## 2.2 Free vibration response: Undamped system

where, $\omega^2 = \frac{k}{m}$

$\therefore$ The general solution of eqn (1) is

$$u(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$
$$= A_1 e^{i\omega t} + A_2 e^{-i\omega t} \quad \text{—— (2)}$$

where $A_1$ & $A_2$ are constant yet unknown.

Using Moivres theorem,

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \Rightarrow e^{ix} + e^{-ix} = 2\cos x \quad \text{—— (3)}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} \Rightarrow e^{ix} - e^{-ix} = 2i\sin x \quad \text{—— (4)}$$

Adding equation (3) & (4)

$$2e^{ix} = 2\cos x + 2i\sin x \Rightarrow \boxed{e^{ix} = \cos x + i\sin x}$$



## 2.2 Free vibration response: Undamped system

Subtracting (4) from (3)

$$2e^{-ix} = 2\cos x - 2i\sin x \Rightarrow \boxed{e^{-ix} = \cos x - i\sin x}$$

$$\therefore \boxed{e^{\pm ix} = \cos x \pm i\sin x}$$

Substituting this value to equation (2)

$$u(t) = A_1(\cos\omega t + i\sin\omega t) + A_2(\cos\omega t - i\sin\omega t)$$
$$= (A_1 + A_2)\cos\omega t + (A_1 - A_2)i\sin\omega t$$

Say $A_1 + A_2 = A$ and $(A_1 - A_2)i = B$ another constant.

$$\therefore \boxed{u(t) = A\cos\omega t + B\sin\omega t.} \quad \text{—— (5)}$$

where, $\omega = \sqrt{\frac{k}{m}}$, natural circular frequency (rad/sec)

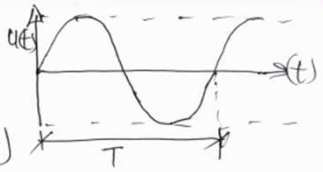
## 2.2 Free vibration response: Undamped system

$T = \frac{2\pi}{\omega}$, Time period

$f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$

natural frequency (cyclic)

Hertz = Hz = cycle/sec.




we have, $u(t) = A\cos\omega t + B\sin\omega t$

or $\dot{u}(t) = -\omega A\sin\omega t + \omega B\cos\omega t$

Initial condition, at $t = 0$, $u(t) = u(0)$ & $\dot{u}(t) = \dot{u}(0)$

$\therefore u(0) = A\cos(0^\circ) + B\sin(0^\circ) = A$

$\therefore u(0) = A$

And $\dot{u}(0) = -\omega A\sin(0^\circ) + \omega B\cos(0^\circ) = \omega B$

$\therefore \dot{u}(0) = B\omega$ and $B = \frac{\dot{u}(0)}{\omega}$

Substituting A & B in eqn (5)

$$u(t) = u(0)\cos\omega t + \frac{\dot{u}(0)}{\omega}\sin\omega t \quad \text{— (6)}$$



## 2.2 Free vibration response: Undamped system (2083/2/19)

Rotating Vector

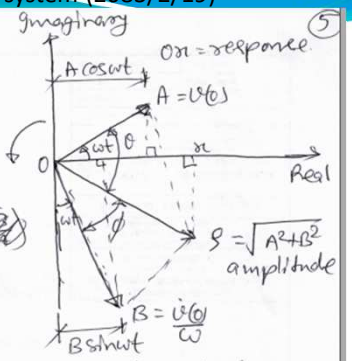
Amplitude of motion

$\rho = \sqrt{[u(0)]^2 + \left[\frac{\dot{u}(0)}{\omega}\right]^2}$ — (7)

Direction: $\tan\theta = \frac{B}{A}$

Phase angle, $\theta = \tan^{-1}\frac{\dot{u}(0)/\omega}{u(0)}$ (8)

$\therefore \theta = \tan^{-1}\frac{\dot{u}(0)}{\omega u(0)}$ — (8)




Phase angle ($\theta$) represents the angular distance by which the resultant motion lags behind the cosine term (real) in the response.

$\tan\phi = \frac{A}{B} = \frac{u(0)\cdot\omega}{\dot{u}(0)} \Rightarrow \therefore \phi = \tan^{-1}\frac{\omega u(0)}{\dot{u}(0)}$ — (9) Phase angle

**∴ Response:**

$u(t) = \rho\cdot\cos(\omega t - \theta)$ — (10)

$u(t) = \rho\cdot\sin(\omega t + \phi)$ — (11)

## 2.2 Free vibration response: Damped system

$M\ddot{u}(t) + c\dot{u}(t) + ku(t) = 0$ … … (12)

We have, $u(t) = e^{st}$

or, $\dot{u}(t) = se^{st}$

or, $\ddot{u}(t) = s^2e^{st}$

$\therefore\ m(s^2e^{st}) + c(se^{st}) + ke^{st} = 0$

Or, $(ms^2 + cs + k)e^{st} = 0$

Since, $e^{st} \neq 0$, $\therefore\ (ms^2 + cs + k) = 0$

Or, $s = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$

Or, $s = \frac{-c}{2m} \pm \sqrt{\frac{c^2}{4m^2} - \frac{k}{m}}$ … … (13)



## 2.2 Free vibration response: Damped system

Since, $\frac{k}{m} = \omega^2$ $\therefore s = \frac{-c}{2m} \pm \sqrt{\frac{c^2}{4m^2} - \omega^2}$ …. …. (14)

Here three types of motion are possible, which are:

$\left(\frac{c}{2m}\right)^2 > \omega^2$ Case I

$\left(\frac{c}{2m}\right)^2 < \omega^2$ Case II

$\left(\frac{c}{2m}\right)^2 = \omega^2$ Case III

## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

Case-III: when $(C/2m)^2 = \omega^2$

Or, $C/2m = \omega$

Or, $C = 2m\omega = C_{cr}$ - Critical Damping value

Or, $Ccr = 2m\sqrt{(k/m)} = 2\sqrt{(mk)} = 2k\sqrt{(m/k)} = 2k/\omega$

$\therefore$ Damping Ratio, $\xi = C/C_{cr} = C/(2m\omega) = C\omega/2k$ ... (15)

And, $s = -C/(2m) = -\omega$

Or, $s = -\omega$

The damping constant 'C' is a measure of the energy dissipated in a cycle of free vibration or in a cycle of forced harmonic vibration.

Damping ratio is the property of the system that also depends on its mass and stiffness.



## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

If $C = C_{cr}$ or, $\xi = 1$, the system returns to equilibrium position without oscillating. (**Critically Damped System**)

If $C > Ccr$ or $\xi > 1$, again the system does not oscillate and returns to its equilibrium position as in the case of $\xi = 1$, but at a slower rate. (**Over Damped System**)

If $C < Ccr$ or $\xi < 1$, the system oscillates its equilibrium position with a progressively decreasing amplitude. (**Under damped System**)

## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

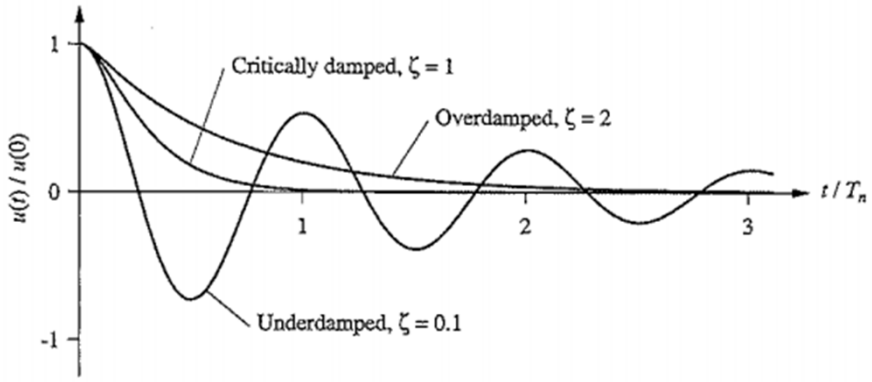



**Figure 2.2.1** Free vibration of underdamped, critically damped, and overdamped systems.



## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

One particular solution is , $u(t) = e^{st}$

Another particular solution is , $u(t) = t.e^{st}$

This solution should not be linearly dependent on the first solution ($e^{st}$).

Because the second particular solution shall not linearly dependent of the first particular solution.

The complete solution is:

$u(t) = A1.e^{st} + A2.t.e^{st} = (A1+t.A2).e^{st}$ ... (16)

Or, $u(t) = (A_1+t.A_2).e^{-\omega t}$

Or, $\acute{u}(t) = (o+A2).\ e^{-\omega t} + (A1+A2).(-\omega).\ e^{-\omega t}$

$= \{A2-\omega(A1+A2)\}\ .\ e^{-\omega t} = \{-\omega A1+A2(1-t\omega)\}\ .\ e^{-wt}$

## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

Initial condition:

at $t = 0$, $v(t) = v(0)$ & $\dot{v}(t) = \dot{v}(0)$

$\therefore v(0) = (A_1 + 0 \cdot A_2) e^{-\omega \cdot 0} = (A_1 + 0) e^{-0}$

$\Rightarrow \boxed{A_1 = v(0)}$

And, $\dot{v}(0) = [-\omega A_1 + A_2(1-0)] e^{-0} = (-\omega A_1 + A_2)$

or, $A_2 = \dot{v}(0) + \omega A_1 = \dot{v}(0) + \omega v(0)$

$\boxed{A_2 = \dot{v}(0) + \omega v(0)}$

Substituting in eqn (18)

$v(t) = \{v(0) + t[\dot{v}(0) + \omega v(0)]\} e^{-\omega t}$

or, $v(t) = [v(0)(1+\omega t) + \dot{v}(0) \cdot t] e^{-\omega t}$ ——— (17)



## 2.2 Critically Damped, Under Damped and Over Damped Systems

The critically damped vibration can also be defined as the smallest amount of damping for which no oscillation occurs in the free vibration response.

Case – I Over Damped System

When $\left(\frac{c}{2m}\right)^2 > \omega^2$ i.e. $\xi = \frac{c}{c_{cr}} > 1$ Or, $C > C_{cr}$

Or, $C > 2m\omega$

Damping Ratio, $\xi = \frac{c}{c_{cr}} = \frac{c}{2m\omega}$

$\therefore S = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \omega^2}$

$= -\xi\omega \pm \sqrt{\xi^2\omega^2 - \omega^2}$

$= -\xi\omega \pm \omega\sqrt{(\xi^2 - 1)}$

Or, $S1 = -\xi\omega + W$, and

$S2 = -\xi\omega - W$

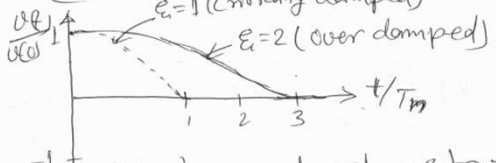
Where, $W = \omega\sqrt{(\xi^2 - 1)}$

## 2.2 Free vibration response: Damped system (Critically Damped, Under

(9)

Complete solution is

$$u(t) = (A_1 \cdot \sinh Wt + B \cosh \cdot Wt) \cdot e^{-\xi_1 \omega t} \quad \text{(18)}$$




Displacement returns to zero in accordance with the exponential decay term.

Case-II : Underdamped system.

when, $\left(\frac{c}{2m}\right)^2 < \omega^2$

or $c < 2m\omega$ or $c < c_{cr}$

$$\xi = \frac{c}{c_{cr}} = \frac{c}{2m\omega} < 1$$

$$\therefore s = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \omega^2}$$



## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

$$\therefore s = -\xi_1\omega \pm \sqrt{\xi^2\omega^2 - \omega^2}$$

$$= -\xi\omega \pm \omega\sqrt{\xi^2 - 1}$$

$$= -\xi\omega \pm \omega\sqrt{-1(1-\xi_1^2)} = -\xi_1\omega \pm \omega\sqrt{i^2(1-\xi^2)}$$

$$= -\xi\omega \pm \omega i\sqrt{(1-\xi^2)}$$

If $\omega_D = \omega\sqrt{(1-\xi^2)}$, damped vibration frequency.

$$\therefore \boxed{s = -\xi_1\omega \pm i\omega_D} \Rightarrow s_1 = -\xi\omega + i\omega_D,\ s_2 = -\xi\omega - i\omega_D$$

Value of $\xi_1$ is very ~~does~~ small. No any construction material have $\xi_1$ greater than 20%. Normally $\xi_1 = 5\%$

Complete solution is:-

$$u(t) = A_1 \cdot e^{s_1 t} + A_2 \cdot e^{s_2 t}$$

$$u(t) = A_1 \cdot e^{(-\xi\omega + i\omega_D)t} + A_2 \cdot e^{(-\xi_1\omega - i\omega_D)t}$$

## 2.2 Free vibration response: Damped system (Critically Damped, Under

or $u(t) = e^{-\xi\omega t}[A_1 e^{i\omega_D t} + A_2 e^{-i\omega_D t}]$ — (19) (10)

or $u(t) = e^{-\xi\omega t}[A\cos\omega_D t + B\sin\omega_D t]$ — (20)

Undamped free vibration,

$u(t) = A_1 e^{i\omega t} + B_2 e^{-i\omega t}$ using Moivre's theorem

we get $u(t) = A\cos\omega t + B\sin\omega t$

or $\dot{u}(t) = -\xi\omega e^{-\xi\omega t}[A\cos\omega_D t + B\sin\omega_D t] + e^{-\xi\omega t}[-A\omega_D\sin\omega_D t + B\omega_D\cos\omega_D t]$

When, $t = 0$, $u(t) = u(0)$ and $\dot{u}(t) = \dot{u}(0)$

$\therefore \dot{u}(0) = -\xi\omega[A + 0] + 1\cdot[-0 + B\omega_D]$

$\dot{u}(0) = -\xi\omega A + B\omega_D$

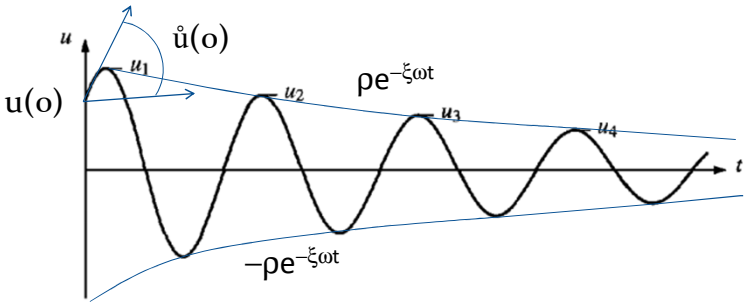
And, $u(0) = 1\cdot[A + 0]$ $\therefore \boxed{A = u(0)}$

and $\dot{u}(0) = -\xi\omega u(0) + B\omega_D$

or $\boxed{B = \frac{1}{\omega_D}[\dot{u}(0) + \xi\omega u(0)]}$



## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)




$u(t) = e^{-\xi\omega t}[u_{(0)}\cdot\cos\omega_D t + 1/\omega_D\{\dot{u}_{(0)} + \xi\omega u_{(0)}\}\sin\omega_D t]$ .. (21)

$\omega_D = \sqrt{(1-\xi^2)}$

If we put $\xi = 0$, $\omega_D = \omega$ and eq$^n$(21) reduces to eq$^n$ of undamped free vibration.

## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

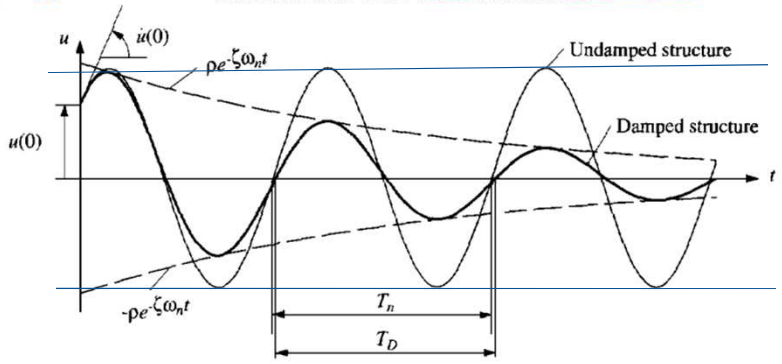



**Figure 2.2.2** Effects of damping on free vibration.

$T_D = 2\pi/\omega_D = 2\pi/\omega\sqrt{(1-\xi^2)} = T/\sqrt{(1-\xi^2)}$

$\omega_D = \omega\sqrt{(1-\xi^2)}$ and $\omega/\omega_D = 1/\sqrt{(1-\xi^2)}$

$T/T_D = \sqrt{(1-\xi^2)} = \omega_D/\omega$, or $T/T_D = \omega_D/\omega$



## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

The displacement amplitude decreases in every cycle of vibration. Eqn (21) indicates that the displacement amplitude decays exponentially with time.

Envelope curve (= $\pm\rho e^{-\xi\omega Dt}$) touches the displacement-time curve at points slightly to the right of its peak value.

Where, $\rho = \sqrt{[\{u_{(0)}\}^2+\{1/\omega_D(\mathring{u}_{(0)}+\xi\omega u_{(0)})\}^2]}$

Damping has the effect:

- Lowering the natural frequency $\omega$ to $\omega_D$, and
- increasing time period from T to $T_D$

## 2.2 Free vibration response: Damped system (Critically Damped, Under

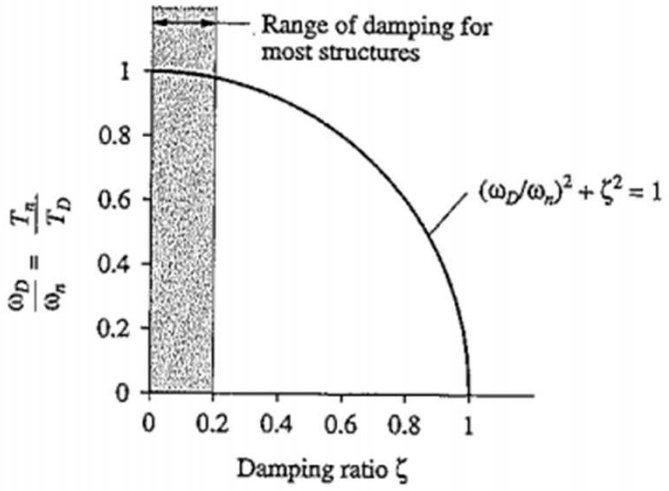



**Figure 2.2.3** Effects of damping on the natural vibration frequency.



## 2.2 Free vibration response: Damped system (Critically Damped, Under Damped and Over Damped Systems)

The more important effect of damping is on the rate at which free vibration decays. This is displayed in Fig. 2.2.4, where the free vibration due to initial displacement $u(0)$ is plotted for four systems having the same natural period $T_n$ but differing damping ratios: $\zeta = 2, 5, 10$, and 20%.

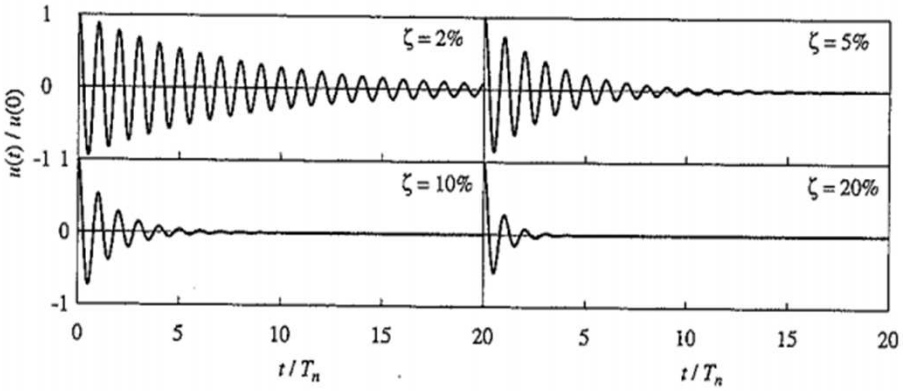



**Figure 2.2.4** Free vibration of systems with four levels of damping: $\zeta = 2, 5, 10$, and 20%.

## 2.3 Logarithmic decrement of response

$$\frac{v(t)}{v(t+T_D)} = \frac{\xi e^{-\xi\omega t}}{\xi e^{-\xi\omega(t+T_D)}} = \frac{e^{-\xi\omega t} \cdot e^{+\xi\omega(t+T_D)}}{e^{-\xi\omega(t+T_D)} \cdot e^{+\xi\omega(t+T_D)}}$$

$$= \frac{e^{-\xi\omega t + \xi\omega t + \xi\omega T_D}}{e^{-\xi\omega t + \xi\omega T_D + \xi\omega t + \xi\omega T_D}} = \frac{e^{\xi\omega T_D}}{e^{0}}$$

$$\therefore \frac{v(t)}{v(t+T_D)} = e^{\xi\omega T_D} = e^{\xi\omega \frac{T}{\sqrt{1-\xi^2}}} = e^{\xi\omega \cdot \frac{2\pi}{\omega\sqrt{1-\xi^2}}} \quad (2$$

$$\therefore \frac{v(t)}{v(t+T_D)} = e^{\left(\frac{2\pi\xi}{\sqrt{1-\xi^2}}\right)} \quad \text{— (23)}$$

$$\therefore \frac{v(t)}{v(t+T_D)} = e^{\left(2\pi\xi \frac{\omega}{\omega_D}\right)} \quad \text{— (24)}$$

For small damping $\xi^2 \simeq 0$ then,

$$\boxed{\frac{v(t)}{v(t+T_D)} = e^{2\pi\xi}} \quad \text{— (25)}$$



## 2.3 Logarithmic decrement of response

Taking log

$$\delta = \ln \frac{v_n}{v_{n+1}} = \frac{2\pi\xi}{\sqrt{1-\xi^2}} \simeq 2\pi\xi \quad \left[\xi^2 \simeq 0\right] \quad \text{— (26)}$$

$$\therefore \frac{v_n}{v_{n+1}} = e^{2\pi\xi} = e^{\delta} \quad \text{— (27)}$$

After n cycles,

$$\frac{v_1}{v_{n+1}} = \frac{v_1}{v_2} \cdot \frac{v_2}{v_3} \cdot \frac{v_3}{v_4} \cdot \cdots \cdot \frac{v_{(n+1)-1}}{v_{(n+1)}}$$

$$= e^{\delta} \cdot e^{\delta} \cdot e^{\delta} \cdots e^{\delta} = e^{n\delta}$$

$$\therefore \boxed{\frac{v(n)}{v(n+m)} = e^{m\delta}} \quad \text{— (28)}$$

## 2.3 Logarithmic decrement of response

$$\delta = \ln \frac{u(t)}{u(t+m)}$$

$$m = \frac{1}{\delta} \ln \frac{u(t)}{u(t+m)} \quad \text{— (29)}$$

After how many cycles would the displacement be reduced to 50%?

$$u(t+m) = 0.5\,u(t)$$

$$\therefore m = \frac{1}{2\pi\xi} * \ln \frac{u(t)}{0.5\,u(t)}$$

$$m = \frac{0.11}{\xi} \quad \text{Normally.}$$

If $\xi = 5\%$ then, $m = 2.2$



## 2.3 Logarithmic decrement of response

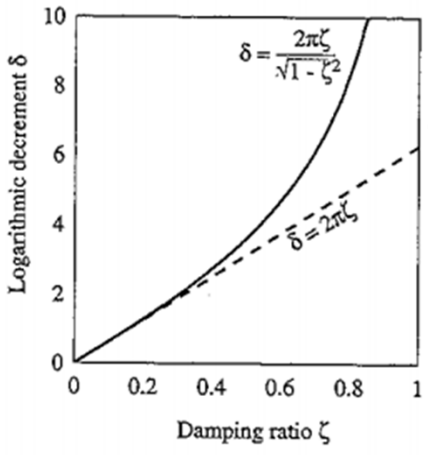



**Figure 2.2.6** Exact and approximate relations between logarithmic decrement and damping ratio.

## 2.3 Logarithmic decrement of response

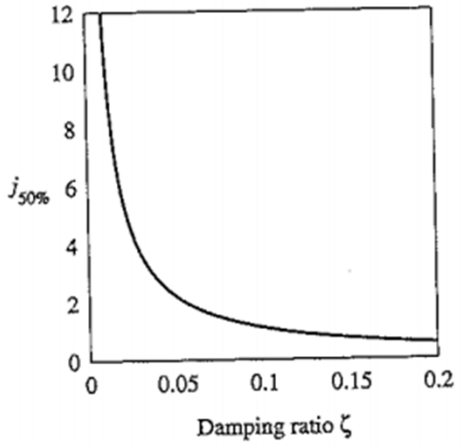



**Figure 2.2.7** Number of cycles required to reduce the free vibration amplitude by 50%.



## 2.3 Logarithmic decrement of response

Q. What would be amplitude after 10 cycles. (1 mark)

$$m = 10 = \frac{1}{2\pi\xi} \cdot \ln \frac{u(n)}{u(n+m)}$$

$$\Rightarrow 20\pi\xi = \ln \frac{u(n)}{u(n+m)}$$

$$\Rightarrow e^{20\pi\xi} = \frac{u(n)}{u(n+m)}$$

$$\Rightarrow u(n+m) = \frac{u(n)}{e^{20\pi\xi}} \quad \text{if } \xi = 5\%$$

$$\therefore u(n+m) = 0.0432\, u(n)$$

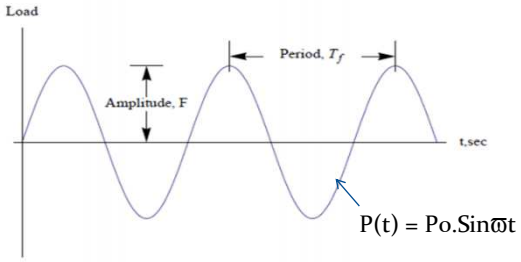
i.e. amplitude would be 0.0432 times the initial amplitude.

## 2.4 Forced vibration response to Harmonic loading: Undamped system

Undamped Forced Vibration Equation of Motion:

$m\ddot{u}(t) + ku(t) = P(t)$ … … (1)

Harmonic Force:




$\varpi$ - Exciting or forcing frequency
$Tp = 2\pi/\varpi$ - Forcing Period



## 2.4 Forced vibration response to Harmonic loading: Undamped system

Undamped system with Harmonic Excitation:

$m\ddot{u}(t) + ku(t) = Po.Sin\varpi t$ … … (2)

General solution of such types of equation is:

$U(t) = u_c(t) + u_p(t)$ … … (3)

- Where, $u_c(t)$ – Complementary Solution
  $u_p(t)$ – Particular solution, depends upon types of load

The response to the harmonic loading can be assumed as harmonic and in phase with loading

## 2.4 Forced vibration response to Harmonic loading: Undamped system

$$v_c(t) = A\cos\omega t + B\sin\omega t \quad \text{——(4)}$$

$v_p(t) = C\cdot\sin\bar{\omega}t$,

C = maximum displacement (amplitude)

$$\dot{v}_p(t) = \bar{\omega} C\cos\bar{\omega}t$$

$$\ddot{v}(t) = -\bar{\omega}^2 \cdot C\sin\bar{\omega}t$$

Substituting in eq$^n$ (3)

$$-m\bar{\omega}^2 C\sin\bar{\omega}t + k\cdot C\sin\bar{\omega}t = P_0\sin\bar{\omega}t$$

$$\Rightarrow (kC - m\bar{\omega}^2 C - P_0)\sin\bar{\omega}t = 0$$

$$\because \sin\bar{\omega}t \neq 0$$

$$\therefore kC - m\bar{\omega}^2 C - P_0 = 0$$

$$\Rightarrow C(k - m\bar{\omega}^2) = P_0$$



## 2.4 Forced vibration response to Harmonic loading: Undamped system

$$\Rightarrow C = \frac{P_0}{(k - m\bar{\omega}^2)} = \frac{P_0}{k\left(1 - \frac{m}{k}\bar{\omega}^2\right)} = \frac{P_0}{k\left(1 - \frac{\bar{\omega}^2}{\omega^2}\right)}$$

$$\left[\because \frac{m}{k} = \omega^2\right]$$

$$\therefore \boxed{C = \frac{P_0}{k\left(1 - \frac{\bar{\omega}^2}{\omega^2}\right)}}$$

Introducing frequency ratio $(\beta) = \frac{\bar{\omega}}{\omega}$

$$\therefore \boxed{C = \frac{P_0}{k(1-\beta^2)}}$$

$$\therefore \boxed{v_p(t) = \frac{P_0}{k(1-\beta^2)}\sin\bar{\omega}t} \quad \text{——(6)}$$

## 2.4 Forced vibration response to Harmonic loading: Undamped system

From eqn (a)

$$u(t) = u_c(t) + u_p(t)$$

$$\therefore u(t) = A\cos\omega t + B\sin\omega t + \frac{P_0}{K(1-\beta^2)}\sin\bar{\omega}t \quad \text{---(7)}$$

Using Initial Condition.

at $t=0$, $u(t) = u(0)$, $\dot{u}(t) = \dot{u}(0)$

$$\therefore u(0) = A + 0 + 0 \Rightarrow A = u(0)$$

$$\text{and } \dot{u}(t) = -\omega A\sin\omega t + \omega B\cos\omega t + \frac{P_0\bar{\omega}}{K(1-\beta^2)}\cos\bar{\omega}t$$

$$\text{at } \dot{u}(0) = 0 + \omega B + \frac{P_0\bar{\omega}}{K(1-\beta^2)}$$



## 2.4 Forced vibration response to Harmonic loading: Undamped system

$$\text{or } B = \frac{1}{\omega}\left[\dot{u}(0) - \frac{P_0\bar{\omega}}{K(1-\beta^2)}\right]$$

$$\boxed{B = \frac{\dot{u}(0)}{\omega} - \frac{P_0\beta}{K(1-\beta^2)}}$$

$u_c(t)$ depends upon the initial condition and not on the type of load, so it is also called Transient Response.

$u_p(t)$ depends upon the type of loading and it is called steady state Response.

$$\therefore u(t) = u(0)\cos\omega t + \left[\frac{\dot{u}(0)}{\omega} - \frac{P_0\beta}{K(1-\beta^2)}\right]\sin\omega t + \frac{P_0}{K(1-\beta^2)}\sin\bar{\omega}t \quad \text{---(8)}$$

$u_c(t)$ [Transient Response] — Steady State Response $u_p(t)$

## 2.4 Forced vibration response to Harmonic loading: Undamped system

If system start from rest

$u(0) = \dot{u}(0) = 0$

$\therefore A = u(0) = 0$ & $B = \frac{\dot{u}(0)}{\omega} - \frac{P_0 \beta}{K(1-\beta^2)} = -\frac{P_0 \beta}{K(1-\beta^2)}$

$u(t) = \frac{P_0}{K(1-\beta^2)} [\sin \bar{\omega} t - \beta \sin \omega t]$ — (9)

Here, $\frac{P_0}{K} = u_{st}$, static displacement

$\frac{1}{1-\beta^2}$ – Magnification factor



## 2.4 Forced vibration response to Harmonic loading: Undamped system

$\frac{P_0}{K} \cdot \frac{1}{1-\beta^2} \Rightarrow$ Static displacement * magnification factor gives the effect of dynamic in Harmonic Excitation.

Response Ratio:

$R(t) = \frac{u(t)}{u_{st}} = \frac{u(t)}{P_0/K} =$ — (10)

$R(t) = \frac{1}{(1-\beta^2)} [\sin \bar{\omega} t - \beta \sin \omega t]$ — (11)

## 2.4 Forced vibration response to Harmonic loading: Undamped system

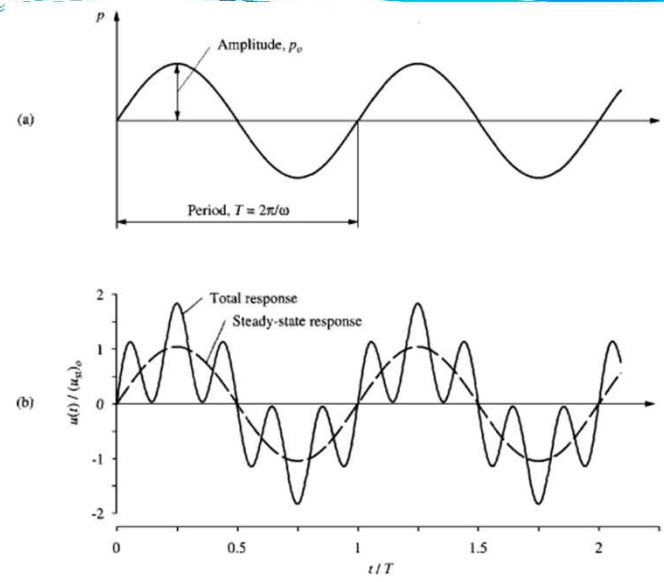



**Figure 3.1.1** (a) Harmonic force; (b) response of undamped system to harmonic force; $\omega/\omega_n = 0.2$, $u(0) = 0$, and $\dot{u}(0) = \omega_n p_o/k$.



## 2.4 Forced vibration response to Harmonic loading: Damped system

Damped system (II)

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = P_o \sin\omega t \quad (12)$$

we know, $\xi = \frac{c}{c_{cr}} = \frac{c}{2m\omega} \Rightarrow \frac{c}{m} = 2\xi\omega$ & $\frac{k}{m} = \omega^2$

Dividing eqn (12) by 'm' we get,

$$\ddot{u}(t) + \frac{c}{m}\dot{u}(t) + \frac{k}{m}u(t) = \frac{P_o}{m}\sin\omega t$$

$$\text{or } \ddot{u}(t) + 2\xi\omega\dot{u}(t) + \omega^2 u(t) = \frac{P_o}{m}\sin\omega t \quad (13)$$

General solution:- $u(t) = u_c(t) + u_p(t)$ (14)

Complementary solution;-

$$u_c(t) = (A\cos\omega_D t + B\sin\omega_D t)\, e^{-\xi\omega t} \quad (15)$$

## 2.4 Forced Harmonic Response (Damped System)

Particular solution: (Amplitude is decreasing)

$$v_p(t) = C \sin\bar{\omega}t + D\cos\bar{\omega}t \quad \text{——— (16)}$$

[In this solution, cosine term is required as well as the sine term, because, in general the response of a damped system is not in phase with the loading.]

$\therefore \dot{v}_p(t) = \bar{\omega} C \cdot \cos\bar{\omega}t - \bar{\omega} D \cdot \sin\bar{\omega}t$

& $\ddot{v}_p(t) = -\bar{\omega}^2 C \cdot \sin\bar{\omega}t - \bar{\omega}^2 D \cdot \cos\bar{\omega}t$

Substituting these value in eqn (13) we get,

$$-\bar{\omega}^2 C \cdot \sin\bar{\omega}t - \bar{\omega}^2 D \cdot \cos\bar{\omega}t + 2\xi\omega[\bar{\omega} C \cdot \cos\bar{\omega}t - \bar{\omega} D \cdot \sin\bar{\omega}t] + \omega^2[C \cdot \sin\bar{\omega}t + D\cos\bar{\omega}t] = \frac{P_0}{m}\sin\bar{\omega}t$$

$$\text{or } [-\bar{\omega}^2 C - 2\xi\omega\bar{\omega} D + \omega^2 C]\sin\bar{\omega}t + [-\bar{\omega}^2 D + 2\xi\omega\bar{\omega} C + \omega^2 D]\cdot \cos\bar{\omega}t = \frac{P_0}{m}\sin\bar{\omega}t$$

Equating coefficient of sine & cosine terms of both sides,



## 2.4 Forced Harmonic Response (Damped System)

$$\therefore -\bar{\omega}^2 C - 2\xi\omega\bar{\omega} D + \omega^2 C = \frac{P_0}{m}$$

$$\text{and } -\bar{\omega}^2 D + 2\xi\omega\bar{\omega} C + \omega^2 D = 0$$

$$\text{or } (\omega^2 - \bar{\omega}^2) D = -2\xi\omega\bar{\omega} C$$

$$\text{or } D = -\frac{2\xi\omega\bar{\omega}}{\omega^2 - \bar{\omega}^2} C$$

$$\text{or } D = -\frac{\frac{2\xi\omega\bar{\omega}}{\omega^2}}{\frac{\omega^2}{\omega^2} - \frac{\bar{\omega}^2}{\omega^2}} C = -\frac{2\xi\beta}{1-\beta^2} C \quad \left[\because \frac{\bar{\omega}}{\omega} = \beta\right]$$

## 2.4 Forced Harmonic Response (Damped System)

$$\therefore \quad -\bar{\omega}^2 C - 2\xi\omega\bar{\omega}\cdot\left(\frac{-2\xi\beta}{1-\beta^2}\right)C + \omega^2 C = \frac{P_0}{m} \qquad (3)$$

$$\text{or}\left[-\bar{\omega}^2 + \omega\bar{\omega}\frac{(2\xi)^2\beta}{(1-\beta^2)} + \omega^2\right]C = \frac{P_0\cdot\omega^2}{K} \quad \left[\because \frac{K}{m} = \omega^2\right]$$

$$\text{or}\left[-\frac{\bar{\omega}^2}{\omega^2} + \frac{\omega\bar{\omega}}{\omega^2}\cdot\frac{(2\xi)^2\beta}{(1-\beta^2)} + \frac{\omega^2}{\omega^2}\right]\cdot C = \frac{P_0}{K}$$

$$\text{or}\left[-\beta^2 + \beta\cdot\frac{(2\xi)^2\beta}{(1-\beta^2)} + 1\right]C = \frac{P_0}{K}$$

$$\therefore \quad C = \frac{P_0}{K}\left[\frac{1}{(1-\beta^2) + \frac{(2\xi\beta)^2}{(1-\beta^2)}}\right]$$

$$\boxed{C = \frac{P_0}{K}\cdot\frac{1-\beta^2}{(1-\beta^2)^2 + (2\xi\beta)^2}} \qquad (17)$$



## 2.4 Forced Harmonic Response (Damped System)

and, $$D = -\frac{2\xi\beta}{(1-\beta^2)}\cdot\frac{P_0}{K}\cdot\frac{(1-\beta^2)}{(1-\beta^2)^2 + (2\xi\beta)^2}$$

$$\boxed{D = \frac{P_0}{K}\cdot\frac{-2\xi\beta}{(1-\beta^2)^2 + (2\xi\beta)^2}} \qquad (18)$$

$\therefore$ Complete solution is.

$$u(t) = (A\cos\omega_d t + B\sin\omega_d t)\cdot e^{-\xi\omega t} + \frac{P_0}{K}\cdot\frac{1-\beta^2}{(1-\beta^2)^2 + (2\xi\beta)^2}\sin\bar{\omega}t$$

$$+ \frac{P_0}{K}\cdot\frac{-2\xi\beta}{(1-\beta^2)^2 + (2\xi\beta)^2}\cdot\cos\bar{\omega}t$$

$$u(t) = (A\cos\omega_d t + B\sin\omega_d t)e^{-\xi\omega t} + \frac{P_0}{K}\cdot\frac{(1-\beta^2)\sin\bar{\omega}t - 2\xi\beta\cos\bar{\omega}t}{(1-\beta^2)^2 + (2\xi\beta)^2}$$

Transient response (affected by initial condition)

Steady state response not affected by initial condition.

(19)

## 2.4 Forced Harmonic Response (Damped System)

Using initial Condition at $t=0$, $u(t) = u(0)$ and $\dot{u}(t) = \dot{u}(0)$ — Not required all

$$\therefore u(0) = (A+0)e^{0} + \frac{\rho_0}{K}\left[\frac{0 - 2\xi\beta}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]$$

$$u(0) = A + \frac{\rho_0}{K}\cdot\frac{-2\xi\beta}{(1-\beta^2)^2 + (2\xi\beta)^2}$$

$$\text{or, } \boxed{A = u(0) + \frac{\rho_0}{K}\cdot\frac{2\xi\beta}{(1-\beta^2)^2 + (2\xi\beta)^2}}$$

and,

$$\dot{u}(t) = (-\omega_D A \sin\omega_D t + \omega_D B\cos\omega_D t)e^{-\xi\omega t} + (A\cos\omega_D t + B\sin\omega_D t)\cdot(-\xi\omega)e^{-\xi\omega t} + \frac{\rho_0}{K}\left[\frac{\bar{\omega}(1-\beta^2)\cos\bar{\omega}t + 2\xi\bar{\omega}\beta\sin\bar{\omega}t}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]$$



## 2.4 Forced Harmonic Response (Damped System)

$$\text{or, } \dot{u}(0) = (0 + \omega_D B)e^{0} + (A+0)(-\xi\omega)e^{0} + \frac{\rho_0}{K}\left[\frac{\bar{\omega}(1-\beta^2) - 0}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]$$

$$\dot{u}(0) = \omega_D B - \xi\omega A + \frac{\rho_0}{K}\left[\frac{\bar{\omega}(1-\beta^2)}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]$$

$$\therefore B = \frac{1}{\omega_D}\left[\dot{u}(0) + \xi\omega\left[u(0) + \frac{\rho_0}{K}\cdot\frac{2\xi\beta}{(1-\beta^2)^2 + (2\xi\beta)^2}\right] + \frac{\rho_0}{K}\left[\frac{\bar{\omega}(1-\beta^2)}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]\right]$$

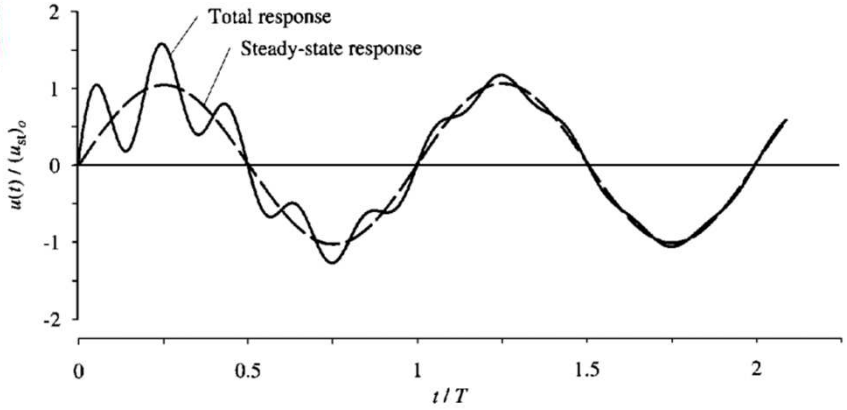



**Figure 3.2.1** Response of damped system to harmonic force; $\omega/\omega_n = 0.2$, $\zeta = 0.05$, $u(0) = 0$, and $\dot{u}(0) = \omega_n p_o/k$.

Transient Response = (Total Response – Steady State Response)

Decay Rate depends on $\varpi/\omega$ and $\xi$.

After totally decay of transient response only force response (Steady State Response) remains.



## 2.4 Forced Harmonic Response (Damped System)

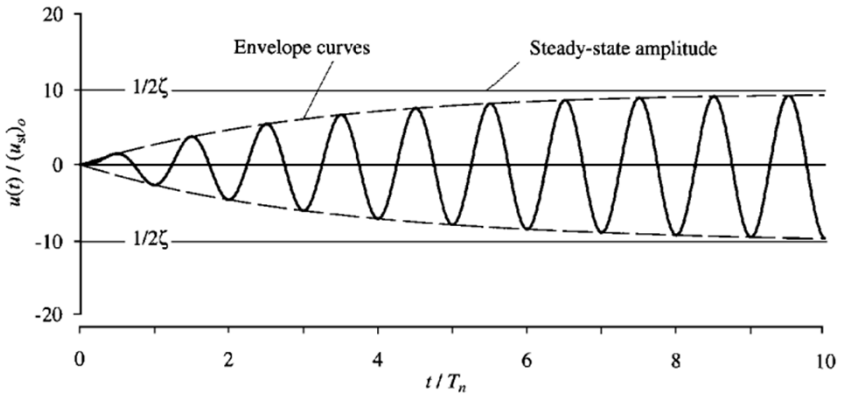



**Figure 3.2.2** Response of damped system with $\zeta = 0.05$ to sinusoidal force of frequency $\omega = \omega_n$; $u(0) = \dot{u}(0) = 0$.

## 2.4 Forced Harmonic Response (Damped System)

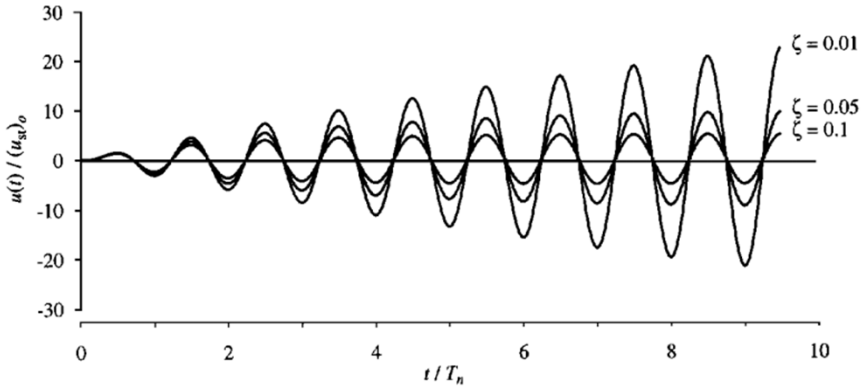
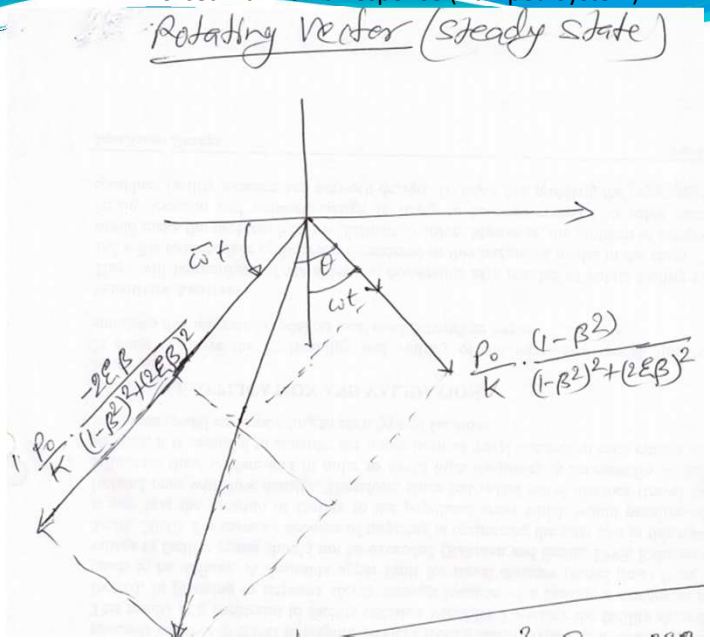



**Figure 3.2.3** Response of three systems—$\zeta = 0.01$, 0.05, and 0.1—to sinusoidal force of frequency $\omega = \omega_n$; $u(0) = \dot{u}(0) = 0$.



## 2.4 Forced Harmonic Response (Damped System)



## 2.4 Forced Harmonic Response (Damped System)

Magnitude $$\rho = \sqrt{\left[\frac{P_0}{K}\frac{(1-\beta^2)}{(1-\beta^2)^2+(2\xi\beta)^2}\right]^2 + \left[\frac{P_0}{K}\cdot\frac{-2\xi\beta}{(1+\beta^2)^2+(2\xi\beta)^2}\right]^2}$$

$$= \frac{P_0}{K}\sqrt{\frac{(1-\beta^2)^2+(2\xi\beta)^2}{\left[(1-\beta^2)^2+(2\xi\beta)^2\right]^2}}$$

$$\text{i.e. } \rho = \frac{P_0}{K}\frac{1}{\sqrt{(1-\beta^2)^2+(2\xi\beta)^2}}$$

Direction, $$\theta = \tan^{-1}\frac{D}{C} = \tan^{-1}\frac{-2\xi\beta}{(1-\beta^2)} \quad \text{— (20)}$$

$$u(t) = u_p(t) = \rho \sin(\bar{\omega}t-\theta) \quad \text{— (21)}$$



## 2.4 Forced Harmonic Response (Damped System)

from eqn (21) (7)

$$u(t) = \rho \sin(\bar{\omega}t-\theta) = \frac{P_0}{K}R_D \sin(\bar{\omega}t-\theta)$$

If $\bar{\omega}/\omega \ll 1$, i.e. $\beta \to 0$ & $R_D = 0 \therefore \rho = \frac{P_0}{K}$

It means that load is very slowly varying or means that it is similar to static loading. For very rigid structure, $\bar{\omega} \cong 0$ so you can perform static analysis, load & response are in phase. ($\phi$ is close to $0°$.)

If $\frac{\bar{\omega}}{\omega} \gg 1$, load is completely out of phase with response. and $\phi$ is close to $180°$ when the force applied to right, displacement would be left.

## 2.4 Forced Harmonic Response (Damped System)

Dynamic magnification factor :-

$$R_D = \frac{\rho}{P_0/K} = \frac{P_0/K \cdot \frac{1}{\sqrt{(1-\beta^2)^2+(2\xi\beta)^2}}}{P_0/K} = \frac{1}{\sqrt{(1-\beta^2)^2+(2\xi\beta)^2}}$$

$$R_D = \frac{1}{\sqrt{(1-\beta^2)^2+(2\xi\beta)^2}} \quad \text{where, } \beta = \frac{\bar{\omega}}{\omega}$$

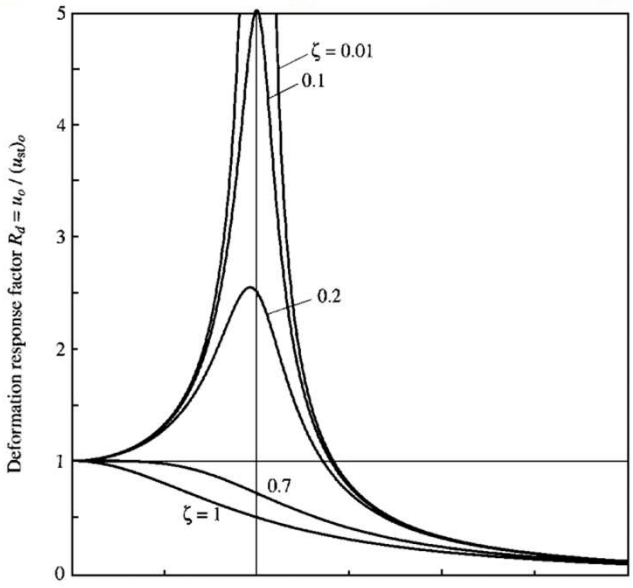
If $\xi = 0$, then, $R_D = \frac{1}{1-\beta^2}$. equal to magnification factor.

$\therefore \rho = \frac{P_0}{K} R_D$ [Dynamic magnification factor when multiplied with $P_0/K$ gives the dynamic response.

(2)



## 2.4 Forced Harmonic Response (Damped System)



## 2.4 Forced Harmonic Response (Damped System)

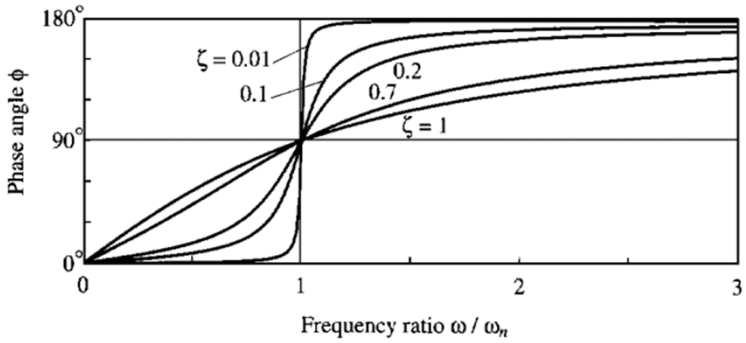



**Figure 3.2.6** Deformation response factor and phase angle for a damped system excited by harmonic force.



## 2.4 Forced Harmonic Response (Damped System)

This result implies that the response is controlled by the damping of the system.

The phase angle $\phi$, which defines the time by which the response lags behind the force, varies with $\omega/\omega_n$ as shown in Fig. 3.2.6. It is examined next for the same three regions of the excitation-frequency scale:

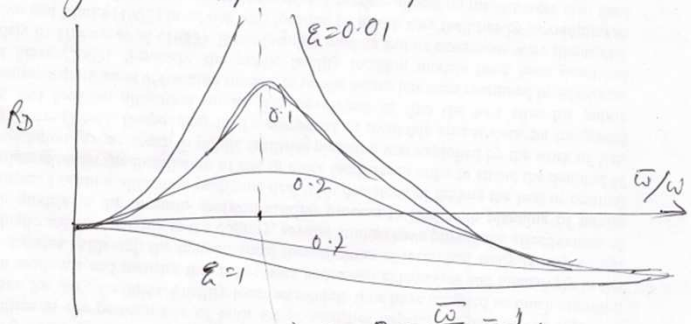
**1.** If $\omega/\omega_n \ll 1$ (i.e., the force is "slowly varying"), $\phi$ is close to 0° and the displacement is essentially in phase with the applied force, as in Fig. 3.2.5a. When the force in Fig. 1.2.1a acts to the right, the system would also be displaced to the right.

**2.** If $\omega/\omega_n \gg 1$ (i.e., the force is "rapidly varying"), $\phi$ is close to 180° and the displacement is essentially out of phase relative to the applied force, as in Fig. 3.2.5c. When the force acts to the right, the system would be displaced to the left.

**3.** If $\omega/\omega_n = 1$ (i.e., the forcing frequency is equal to the natural frequency), $\phi = 90°$ for all values of $\zeta$, and the displacement attains its peaks when the force passes through zeros, as in Fig. 3.2.5b.

Resonant Frequencies and Resonant Response

Resonant frequency = forcing frequency at which the largest response amplitude occurs.




i.e. at $\bar{\omega} = \omega$ i.e. $\beta = \frac{\bar{\omega}}{\omega} = 1$.





i.e. at $\bar{\omega} = \omega$ i.e. $\beta = \frac{\bar{\omega}}{\omega} = 1$.

Undamped ($\xi = 0$)

$$v_p(t) = \frac{P_0}{K} \frac{(1-\beta^2)\sin\bar{\omega}t - 2\xi\beta\cos\bar{\omega}t}{(1-\beta^2)^2 + (2\xi\beta)^2}$$

$$= \frac{P_0}{K} \cdot \frac{\sin\bar{\omega}t}{(1-\beta^2)}$$

If $\beta = 1$, $v_p(t)$ = indeterminate.

then, $v_p(t) \neq C\sin\bar{\omega}t$

$\therefore v_p(t) = Ct\cos\bar{\omega}t$

$\dot{v}_p(t) = C\cos\bar{\omega}t - Ct\bar{\omega}\sin\bar{\omega}t$

$\ddot{v}_p(t) = -C\bar{\omega}\sin\bar{\omega}t - [C\bar{\omega}\sin\bar{\omega}t + Ct\bar{\omega}^2\cos\bar{\omega}t]$

## 2.4 Forced Harmonic Response (Damped System)

Or, $\ddot{v}_p(t) = -c\bar{\omega}\sin\bar{\omega}t - c\bar{\omega}[\bar{\omega}t\cos\bar{\omega}t + \sin\bar{\omega}t]$

$= -\bar{\omega}c\sin\bar{\omega}t - c\bar{\omega}^2 t\cos\bar{\omega}t - c\bar{\omega}\sin\bar{\omega}t$

Substituting in eqn of motion:

$-m\bar{\omega}c\sin\bar{\omega}t - mc\bar{\omega}^2 t\cos\bar{\omega}t - mc\bar{\omega}\sin\bar{\omega}t + kct\cos\bar{\omega}t = P_0\sin\bar{\omega}t$

$-2\bar{\omega}c\sin\bar{\omega}t - c\bar{\omega}^2 t\cos\bar{\omega}t + \frac{kc}{m}t\cos\bar{\omega}t = \frac{P_0}{m}\sin\bar{\omega}t$

$-2\bar{\omega}c\sin\bar{\omega}t - c\bar{\omega}^2 t\cos\bar{\omega}t + \bar{\omega}^2 ct\cos\bar{\omega}t = \frac{P_0}{m}\sin\bar{\omega}t$

$-\frac{2c\sin\bar{\omega}t}{\bar{\omega}} - ct\cos\bar{\omega}t + ct\cos\bar{\omega}t = \frac{P_0}{m\bar{\omega}^2}\sin\bar{\omega}t$

$-\frac{2c\sin\bar{\omega}t}{\bar{\omega}} = \frac{P_0}{k}\sin\bar{\omega}t$



## 2.4 Forced Harmonic Response (Damped System)

$-\frac{2c\sin\bar{\omega}t}{\bar{\omega}} = \frac{P_0}{k}\sin\bar{\omega}t$

$-\frac{2c}{\bar{\omega}} = \frac{P_0}{k}$

$\therefore c = -\frac{P_0}{2k}\bar{\omega} \quad \because \omega = \bar{\omega}$

$\therefore \boxed{c = -\frac{P_0}{2k}\omega}$

$\boxed{v_p(t) = -\frac{P_0}{2k}\omega t\cos\bar{\omega}t}$

Complete solution:

$v(t) = A\cos\omega t + B\sin\omega t - \frac{P_0}{2k}\omega t\cos\bar{\omega}t$

Dampeel

$$v_p(t) = \frac{P_0}{K} \cdot \frac{(1-\beta^2)\sin\bar{\omega}t - 2\xi\beta\cos\bar{\omega}t}{(1-\beta^2)^2 + (2\xi\beta)^2}$$

when, $\bar{\omega} = \omega$, $\beta = 1$

$$v_p(t) = \frac{P_0}{K} \times \left[\frac{0 - 2\xi\cos\bar{\omega}t}{0 + (2\xi)^2}\right]$$

$$= \frac{P_0}{K} \cdot \frac{-2\xi\cos\bar{\omega}t}{(2\xi)^2} \quad \because \bar{\omega} = \omega$$

$$\therefore v_p(t) = -\frac{P_0}{K} \cdot \frac{\cos\omega t}{2\xi}$$

Complete solution,

$$v(t) = (A\cos\omega_D t + B\sin\omega_D t)\, e^{-\xi\omega t} - \frac{P_0}{K}\frac{\cos\omega t}{2\xi}$$



# THANK YOU

## STRUCTURAL DYNAMICS

**2.0 Single Degree of Freedom (SDOF) System**

2.5 Energy Dissipated by Damping (viscous damping)
2.6 Forced Vibration Response to Periodic Loading: Undamped system (Transformation of loading and response using Fourier series)
2.7 Forced Vibration Response to Impulsive load (half-sine wave pulse, rectangular pulse, ramp pulse, triangular pulse)
2.8 Forced Vibration Response to General Dynamic Loading: Unit-impluse and Dirac-delta function, Unit-impulse response function, Duhamel integral and convolution integral, Numerical evaluation of Duhamel integral
2.9 Vibration measuring instruments (Displacement meter and accelerometer)
2.10 Vibration isolation (Vertical oscillating force, harmonic motion of the base, force transmissibility, unbalanced mass amplitude of rotating unbalanced system)
2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect



## 2.5 Energy Dissipated by Damping

Consider the steady state motion of an SDOF system having,
$P(t) = P0.\text{Sin}\varpi t$

The energy dissipated by viscous damping in one cycle of harmonic vibration is

$$E_d = \int f_D . du = \int_0^{2\pi/\varpi} (C.\dot{u})(\dot{u}dt) = \int_0^{2\pi/\varpi} (C.\dot{u}^2)dt$$

$$= C\int_0^{2\pi/\varpi} [\varpi\rho.\text{Cos}(\varpi t - \theta)]^2 dt$$

Since $u(t) = \rho.\text{Sin}(\varpi t-\theta)$ and $\dot{u}(t) = \varpi\rho.\text{Cos}(\varpi t-\theta)$

$$\therefore E_d = C.[\varpi^2\rho^2\frac{\pi}{\varpi}] = \pi C\varpi\rho^2 = \pi(2m\omega\xi)\varpi\rho^2 = 2\pi\xi.\frac{k}{\omega}.\varpi\rho^2$$

Since, $C = 2m\omega\xi$ and $m\omega = \frac{k}{\varpi}$

$$\therefore E_d = 2\pi\xi\beta\rho^2 k \quad \ldots \quad \ldots \quad (8)$$

Energy dissipated is proportional to the square of amplitude of the motion.

## 2.5 Energy Dissipated by Damping

It is not a constant value for any given amount of damping and amplitude of excitation since the energy dissipated increases linearly with excitation frequency.
In steady-state vibration, the energy input to the system due to applied force is dissipated in viscous damping.

$E_I = \int P(t)\text{du}$

$= \int_0^{2\pi/\omega} P(t)\dot{u}dt = \int_0^{2\pi/\omega} (Po.Sin\varpi t)(\omega\rho Cos(\varpi t - \theta))dt$

$= \pi.Po.\rho.Sin\theta$

But, $\theta = \tan^{-1}\frac{2\xi\beta}{(1-\beta^2)}$, $\tan\theta = \frac{2\xi\beta}{(1-\beta^2)}$



## 2.5 Energy Dissipated by Damping

$\therefore Sin\theta = \frac{\frac{2\xi\beta}{(1-\beta^2)}}{\left[\frac{2\xi\beta}{(1-\beta^2)}\right]^2 + 1} = \frac{2\xi\beta}{(2\xi\beta)^2 + (1-\beta^2)^2}$

$\therefore E_I = 2\pi\xi\beta.k.\rho^2$ ... ... (9)

$\therefore$ From Equations (8) and (9),

$\therefore E_I = E_d$

Over each cycle of harmonic vibration the changes in potential energy equal to the strain energy of the spring and kinetic energy are zero.

$E_s = \int f_s du = \int_0^{2\pi/\omega} (ku).\dot{u}dt = \int_0^{2\pi/\omega} k[\rho Sin(\varpi t - \theta)][\varpi\rho Cos(\varpi t - \theta)]dt$

$= k\rho^2\varpi \int_0^{2\pi/\omega} Sin(\varpi t - \theta).Cos(\varpi t - \theta)dt = 0$

## 2.5 Energy Dissipated by Damping

$$E_k = \int f_I du = \int_0^{2\pi/\omega} (m\ddot{u}).\dot{u}dt$$

$= \int_0^{2\pi/\omega} m[-\varpi^2 \rho.\text{Sin}(\varpi t-\theta).[\varpi\rho.\text{Cos}(\varpi t-\theta)dt$

$= -m\varpi^3\rho^2 \int_0^{2\pi/\omega} \text{Sin}(\varpi t-\theta).\text{Cos}(\varpi t-\theta)dt = 0$



## 2.6 Forced Vibration Response to Periodic Forces

A periodic excitation implies that the excitation has been in existence for a long time, by which time the transient response associated with the initial displacement and velocity has decayed.

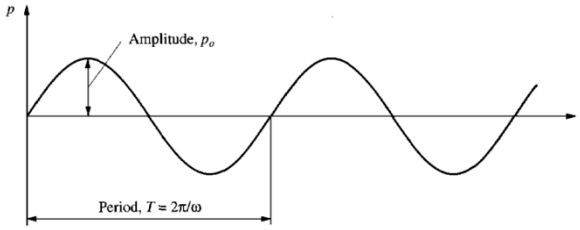
i.e. after some time only steady state response exist.

Periodic function: $f(x) = f(x+P)$

Where, P is constant, called fundamental period

$\therefore$ Periodic loading, $P(t) = P(t+nT)$ … … (1)

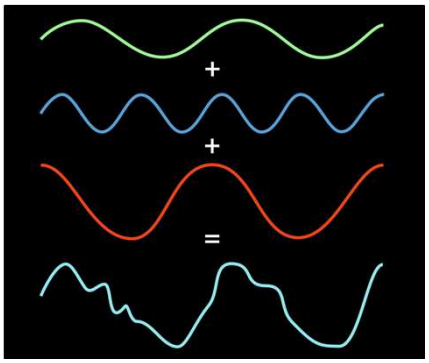
(in general for all positive integers)



## 2.6 Forced Vibration Response to Periodic Forces

Fourier Series:

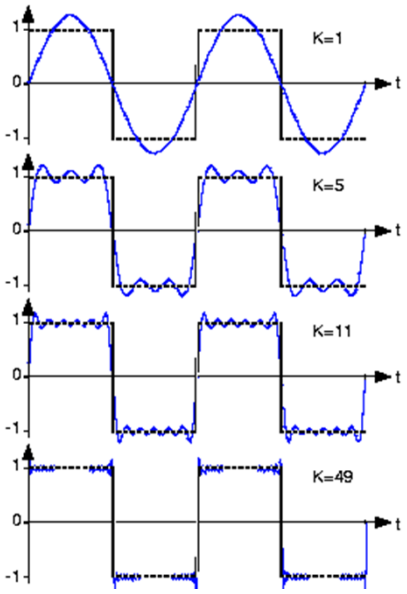
A Fourier series is an expansion of a periodic function in terms of an infinite sum of Sine and Cosines. Fourier series make use of the orthogonality relationships of the Sine and Cosine functions






## 2.6 Forced Vibration Response to Periodic Forces

Square wave




After adding infinite numbers of sine series you got square wave

## 2.6 Forced Vibration Response to Periodic Forces



## 2.6 Forced Vibration Response to Periodic Forces

*Or,*

$$P(t) = a_0 + \sum_0^\infty a_n Cos\left(\frac{2n\pi}{T_p}t\right) + \sum_0^\infty b_n Sin\left(\frac{2n\pi}{T_p}t\right) \ldots \quad \ldots \quad (4)$$

Where harmonic amplitude coefficients are,

$$a_0 = \frac{1}{L}\int_{-L}^{+L} P_{(t)}dt = \frac{1}{T_p}\int_0^{T_p} P_{(t)}dt \quad \ldots. \quad \ldots. \quad (5)$$

$$a_n = \frac{1}{L}\int_{-L}^{L} P(t)\cos\frac{n\pi t}{L}dt = \frac{2}{T_p}\int_0^{T_p} P(t)\cos\frac{2n\pi}{T_p}t\,dt$$

$$a_n = \frac{2}{T_p}\int_0^{T_p} P(t)\cdot\cos n\omega t\cdot dt \quad \text{——} (6)$$

$$b_n = \frac{1}{L}\int_{-L}^{L} P(t)\sin\frac{n\pi t}{L}dt = \frac{2}{T_p}\int_0^{T_p} P(t)\sin\frac{2n\pi}{T_p}t\,dt$$

## 2.6 Forced Vibration Response to Periodic Forces

$$= \frac{2}{T_p}\int_0^{T_p} P(t)\sin n\bar{\omega}t\,dt \quad \text{—— (7)}$$

Alternatively eqn (2) can be simplified as:

$$P(t) = P_0 + \sum_{n=}^{\infty} P_n \cos(n\bar{\omega}t - \phi_n) \quad \text{—— (8)}$$

where, $P_0 = a_0$ and the amplitude of the nth harmonics

$$P_n = \sqrt{a_n^2 + b_n^2} \quad \text{—— (9)}$$

Phase angle, $\phi_n = \tan^{-1}\left(\frac{b_n}{a_n}\right)$ —— (10)



## 2.6 Forced Vibr

Fourier's series:-

$$P(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\bar{\omega}_n t + \sum_{n=1}^{\infty} b_n \sin\bar{\omega}_n t \quad \text{(2)}$$

(9)

### Response to Fourier series loading

### Undamped :-

The steady-state response produced in undamped SDOF by each ~~sine~~ sine term is given by:

$$v_n(t) = \frac{b_n}{K}\cdot\frac{1}{(1-\beta_n^2)}\sin n\bar{\omega}t \quad \text{—— (11)}$$

where, $\beta_n = \frac{\bar{\omega}_n}{\omega} = \frac{n\bar{\omega}}{\omega} = n\beta$

Similarly the steady-state response to each cosine term is given by,

$$v_n(t) = \frac{a_n}{K}\cdot\frac{1}{(1-\beta_n^2)}\cos n\bar{\omega}t \quad \text{— (12)}$$

## 2.6 Forced Vibration Response to Periodic Forces

Fourier's series:-

$$P(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos \bar{\omega} n t + \sum_{n=1}^{\infty} b_n \sin \bar{\omega} n t \quad \text{--- (2)}$$

And the steady-state response to the constant loading,

$$v_0 = \frac{a_0}{K} \quad \text{--- (13)}$$

∴ The total periodic response to undamped SDOF system (steady-state):

$$v(t) = \frac{1}{K}\left[a_0 + \sum_{n=1}^{\infty} \frac{a_n}{(1-\beta_n^2)} \cdot \cos n\bar{\omega} t + \sum_{n=1}^{\infty} \frac{b_n}{(1-\beta_n^2)} \sin n\bar{\omega} t\right] \quad \text{--- (13)}$$

$$\text{or } v(t) = \frac{1}{K}\left[a_0 + \sum_{n=1}^{\infty} \frac{1}{(1-\beta_n^2)} \left[a_n \cos n\bar{\omega} t + b_n \sin n\bar{\omega} t\right]\right] \quad \text{--- (14)}$$



## 2.6 Forced Vibration Response to Periodic Forces

Damped system (SDOF) (10')

Steady state response by each sine terms:

$$v_n(t) = \frac{b_n}{K} \cdot \frac{1}{(1-\beta_n^2)^2 + (2\xi\beta_n)^2} \times \left[(1-\beta_n^2)\sin \bar{\omega} n t - 2\xi\beta_n \cos \bar{\omega} n t\right] \quad \text{--- (15)}$$

Due to cosine terms:

$$v_n(t) = \frac{a_n}{K} \cdot \frac{1}{(1-\beta_n^2)^2 + (2\xi\beta_n)^2} \cdot \left[2\xi\beta_n \sin \bar{\omega} n t + (1-\beta_n^2)\cos \bar{\omega} n t\right] \quad \text{--- (16)}$$

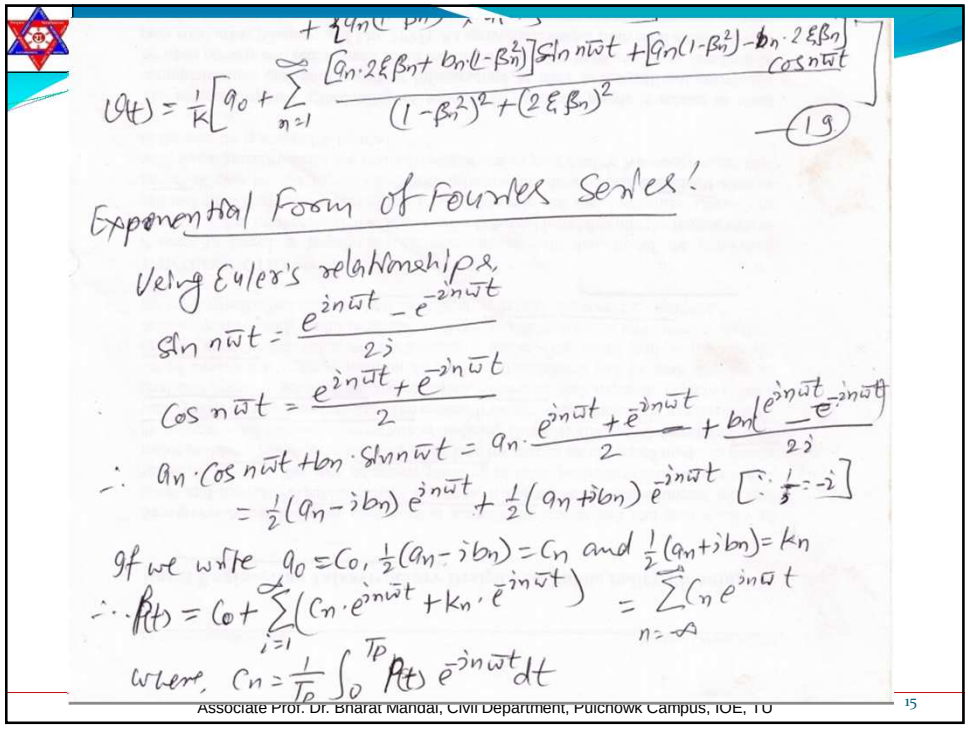
Due to constant term:

$$v_0 = \frac{a_0}{K} \quad \text{--- (17)}$$

Total steady state response:

∴ Total steady state response:

$$v(t) = \frac{1}{K}\left[a_0 + \sum_{n=1}^{\infty} \frac{1}{(1-\beta_n^2)^2 + (2\xi\beta_n)^2} \left[\{a_n \cdot 2\xi\beta_n + b_n(1-\beta_n^2)\}\sin \bar{\omega} n t + \{a_n(1-\beta_n^2) - b_n 2\xi\beta_n\}\cos \bar{\omega} n t\right]\right] \quad \text{--- (18)}$$



Exponential Form of Fourier Series:

Using Euler's relationships,

$$\sin n\bar{\omega}t = \frac{e^{jn\bar{\omega}t} - e^{-jn\bar{\omega}t}}{2j}$$

$$\cos n\bar{\omega}t = \frac{e^{jn\bar{\omega}t} + e^{-jn\bar{\omega}t}}{2}$$

$$\therefore a_n \cdot \cos n\bar{\omega}t + b_n \cdot \sin n\bar{\omega}t = a_n \cdot \frac{e^{jn\bar{\omega}t} + e^{-jn\bar{\omega}t}}{2} + b_n \frac{e^{jn\bar{\omega}t} - e^{-jn\bar{\omega}t}}{2j}$$

$$= \frac{1}{2}(a_n - jb_n) e^{jn\bar{\omega}t} + \frac{1}{2}(a_n + jb_n) e^{-jn\bar{\omega}t} \quad \left[\because \frac{1}{j} = -j\right]$$

If we write $a_0 = C_0$, $\frac{1}{2}(a_n - jb_n) = C_n$ and $\frac{1}{2}(a_n + jb_n) = k_n$

$$\therefore P(t) = C_0 + \sum_{n=1}^{\infty} (C_n \cdot e^{jn\bar{\omega}t} + k_n \cdot e^{-jn\bar{\omega}t}) = \sum_{n=-\infty}^{\infty} C_n e^{jn\bar{\omega}t}$$

where, $C_n = \frac{1}{T_p} \int_0^{T_p} P(t)\, e^{-jn\bar{\omega}t}\, dt$





Where, Kn is the complex conjugate of Cn, formed as for any complex number by reversing the sign of the imaginary part.

Now since terms with negative "n" are complex conjugates of terms with positive "n", then

$$\sum_{n=1}^{\infty} k_n e^{-in\varpi t} = \sum_{n=-1}^{-\infty} C_n e^{in\varpi t}$$

$\therefore$ P(t) = $C_0 + \sum_{n=1}^{\infty} C_n e^{in\varpi t} + \sum_{n=-1}^{-\infty} C_n e^{in\varpi t}$

Noting the nontrivial relationship, $C_0 = C_0 e^{in\varpi t}$, for n = 0

P(t) = $\sum_{-\infty}^{\infty} C_n e^{in\varpi t}$

## 2.6 Forced Vibration Response to Periodic Forces

$$P(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\bar{\omega}t}$$

where, $C_n = \frac{1}{T_p}\int_{-T_p/2}^{T_p/2} P(t)\, e^{-in\bar{\omega}t}\, dt \quad [n = 0, \pm 1, \pm 2, \cdots]$

$$\therefore m\ddot{u}(t) + c\dot{u}(t) + ku(t) = P(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\bar{\omega}t}$$

Steady state solution: (Clough - P98)

$$u(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(i\bar{\omega})\, P(i\bar{\omega})\, e^{i\bar{\omega}t}\, dt$$

$$H(i\bar{\omega}) = \frac{1}{k}\left[\frac{1}{(1-\beta^2) + i(2\xi\beta)}\right] = \frac{1}{k}\left[\frac{(1-\beta^2) - i(2\xi\beta)}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]$$

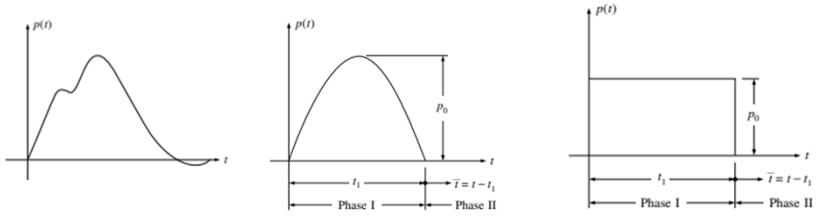


## 2.7 Forced Vibration Response to Impulsive Forces

Forced Vibration Response to Impulsive Force:

**Impulsive** Force:

- Single Principal Impulse
- Relatively short duration
- Damping has much less importance
- No time for steady state response
- Vibration take palce according to natural frequency (free vibration)



## 2.7 Forced Vibration Response to Impulsive Forces

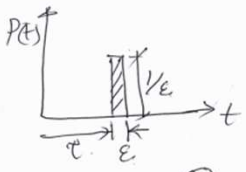
Response to unit impulse:-

Force, $P(t) = \frac{1}{\varepsilon}$

Time duration $= \varepsilon$

Starting at time instant $t = \tau$

As $\varepsilon \to 0$ $P(t) \to \infty$




Real magnitude of impulse $= \int P(t)\,dt = 1$ — (1) [called unit impulse]

Dirac delta Function $\delta(t-\tau)$ mathematically defines a unit impulse centered at $t = \tau$.

$\therefore \int_0^\infty \delta(t-\tau)\,dt = 1, \quad 0 < \tau < \infty$ — (2)

when $t \neq \tau$ $\delta(t-\tau) = 0$ — (3)

$\therefore \int_0^\infty P(t)\cdot\delta(t-\tau)\,dt = P(\tau), \quad 0 < \tau < \infty$ — (4)



## 2.7 Forced Vibration Response to Impulsive Forces

(2)

Newton's II law,

$\frac{d}{dt}(m\dot{x}) = p \Rightarrow m\ddot{x} = p$ — (5)

or $p = m\ddot{x}$ — (6)

Integrating both sides, w.r.t. $t$,

$\int_{t_1}^{t_2} p\,dt = \int_{t_1}^{t_2} m\ddot{x}\,dt = m(\dot{x}_2 - \dot{x}_1) = m\Delta\dot{x}$ — (7)

$\therefore$ magnitude of impulse $\equiv$ change in momentum.

If force acts for infinitesimally short duration so that the spring and damper have no time to respond. Thus a unit impulse at $t = \tau$ imparts to the mass, $m$ the velocity, [$\dot{x}_1 = 0$ & $\dot{x}_2 = \dot{x}(\tau)$ at $t = \tau$]

$\dot{x}(\tau) = \frac{1}{m}$ — (8)

[for $t$ to and up to

## 2.7 Forced Vibration Response to Impulsive Forces

$v(\tau) = m$

But the displacement is zero prior to and up to the impulse.

$\therefore v(\tau) = 0$ ——— (9)

∴ A unit impulse causes free vibration of SDOF system due to the initial displacement and velocity given by eqn (8) & (9) respectively.

In general, Equation of motion is,

$v(t) = A\cos\omega t + B\sin\omega t$

$v(t) = v(0)\cos\omega t + \frac{\dot{v}(0)}{\omega}\sin\omega t$

∴ Displacement start at time $\tau$, $v(0) \to v(\tau)$ and $\dot{v}(0) \to \dot{v}(\tau)$

$v(t) = v(\tau)\cos\omega(t-\tau) + \frac{\dot{v}(\tau)}{\omega}\sin\omega(t-\tau)$

when $t \to \tau$, $(t-\tau) = 0$



## 2.7 Forced Vibration Response to Impulsive Forces

$v(t \to \tau) = v(t-\tau=0)\cdot\cos\omega(t-\tau) + \frac{\dot{v}(t-\tau=0)}{\omega}\sin\omega(t-\tau)$ (3)

$h(t-\tau) \Leftrightarrow v(t-\tau)$ [Here ~~it is not~~ $v$ is not change in $v$ w.r.t $dt$ but it represents time-history response to the differential impulse over $t \geq \tau$

$v(t) \doteq h(t-\tau) = 0\cdot\cos\omega(t-\tau) + \frac{1}{m}\cdot\frac{1}{\omega}\sin\omega(t-\tau)$

or $\boxed{h(t-\tau) = \frac{1}{m\omega}\sin\omega(t-\tau), \quad t \geq \tau}$ ——— (10)

where, $h(t-\tau)$ – unit impulse response function.

For damped system

$v(t) = e^{-\xi\omega t}(A\cos\omega_D t + B\sin\omega_D t)$

$= e^{-\xi\omega t}\left[v(0)\cos\omega_D t + \frac{\dot{v}(0) + v(0)\omega\xi}{\omega_D}\cdot\sin\omega_D t\right]$

## 2.7 Forced Vibration Response to Impulsive Forces

$$\therefore \left[ h(t-\tau) = \frac{1}{m\omega_D} \cdot e^{-\xi\omega(t-\tau)} \cdot \sin\omega_D(t-\tau),\ t \geq \tau \quad \text{---} (11) \right]$$

If Impulse is other than unity, then,

$$u(t) = I \cdot h(t-\tau) \quad \text{---} (12)$$ ~~undamped~~



## 2.7 Forced Vibration Response to Impulsive Forces

<u>Response to arbitrary Force</u>

A force P(t) varying arbitrary with time can be represented as a sequence of infinitesimally short impulses. The response of a linear dynamic system to one of these impulse, the one at time $\tau$ of magnitude $P(\tau)d\tau$ is this magnitude times the unit impulse response.

i.e. $du(\tau) = [P(\tau)d\tau].h(t-\tau)$ $\quad t > \tau \quad \ldots \quad \ldots (13)$

The response of the system at time t is the sum of responses to all impulses up to that time. Thus,

$u(t) = \int_0^t P(\tau).h(t-\tau)d\tau \quad \ldots. \quad \ldots. \quad (14)$

This is known as the Convolution integral a general result that applies to any linear dynamic system.

Substituting eqn (10) & (11) gives

Duhamel's integral:

$$u(t) = \frac{1}{m\omega}\int_0^t p(\tau)\sin[\omega(t-\tau)]\,d\tau \quad \text{——— (15) undamped}$$

$$u(t) = \frac{1}{m\omega_D}\int_0^t p(\tau)\, e^{-\xi\omega(t-\tau)}\sin[\omega_D(t-\tau)]\,d\tau \quad \text{— (16) damped}$$

Implicit in this result are at rest initial conditions.
i.e. $u(0)=0$ & $\dot{u}(0)=0$

System is linear so superposition apply.
(not apply for non-linear system)

If $p(\tau)$ is a simple function, closed form evaluation of the integral is possible.

i.e. Duhamel's integral method is an alternative solution and in case of complicated function, numerical methods applies.



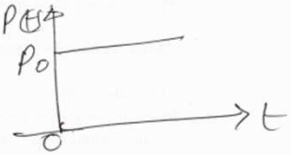


(5)

Response to step and ramp forces:

Step Force:

A step force jumps suddenly from zero to $P_0$ and stays constant.




Undamped:

$$P(t) = P_0 \quad \text{——— (17)}$$

from eqn (10)

$$h(t-\tau) = \frac{1}{m\omega}\sin\omega(t-\tau), \quad t \geq \tau$$

$$\omega^2 = \frac{k}{m}$$

from eqn (15)

$$u(t) = \frac{1}{m\omega}\int_0^t p(\tau)\sin\omega(t-\tau)\,d\tau$$

$$= \frac{P_0}{m\omega}\int_0^t \sin\omega(t-\tau)\,d\tau = \frac{P_0}{m\omega^2}(1-\cos\omega t)$$

$$v(t) = \frac{P_0}{K}(1-\cos\omega t) = \frac{P_0}{K}\left(1-\cos\frac{2\pi t}{T}\right) \quad \text{—(18)}$$

For max value

$$\frac{dv(t)}{dt} = 0 \quad \dot{v}(t) = 0$$

$$\text{or } 0 = \frac{P_0}{K}(0 - \omega\sin\omega t) = \frac{P_0}{K}\omega\sin\omega t$$

$$\text{or } \sin\omega t = 0 \Rightarrow \omega t_0 = n\pi$$

$$\text{or } t_0 = \frac{n\pi}{\omega} = \frac{n}{2}\cdot\frac{2\pi}{\omega} = \frac{n}{2}T$$

$$v_0 = v_{max} = \frac{P_0}{K}\left(1-\cos\frac{2\pi}{T}\cdot\frac{nT}{2}\right) = \frac{P_0}{K}(1-\cos n\pi)$$

when $n$ = odd, $\cos n\pi = -1$

$$\therefore v_{max} = 2\frac{P_0}{K}, \quad \text{but } \frac{P_0}{K} = v_{st}, \text{ static displacement}$$

—(19)





Damped (6)

$$h(t-\tau) = \frac{1}{m\omega_D}e^{-\xi\omega(t-\tau)}\cdot\sin\omega_D(t-\tau), \quad t \geq \tau$$

and,

$$v(t) = \frac{1}{m\omega_D}\int P(\tau)\cdot e^{-\xi\omega(t-\tau)}\cdot\sin\omega_D(t-\tau)\,d\tau$$

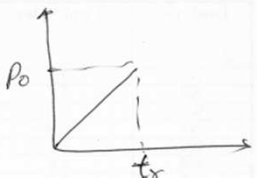
$$= \frac{P_0}{m\omega_D}\int e^{-\xi\omega(t-\tau)}\cdot\sin\omega_D(t-\tau)\,d\tau$$

$$v(t) = \frac{P_0}{K}\left[1 - e^{-\xi\omega(t-\tau)}\left(\cos\omega_D t + \frac{\xi}{\sqrt{1-\xi^2}}\sin\omega_D t\right)\right] \quad \text{—(20)}$$

classical method may be easier.

Response to Ramp Force

$$P(t) = \frac{P_0}{t_r}t \quad \text{—(21)}$$




Using Duhamel integral:-

$$u(t) = \frac{1}{m\omega}\int_0^t P(\tau)\cdot \sin\omega(t-\tau)\,d\tau$$

$$= \frac{P_0}{m\omega t_r}\int_0^t \tau \sin\omega(t-\tau)\,d\tau$$

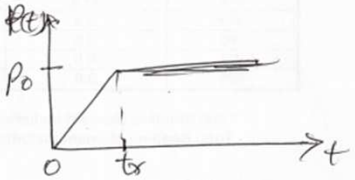
$$\therefore u(t) = \frac{P_0}{K}\left[\frac{t}{t_r} - \frac{\sin\omega t}{\omega t_r}\right] \quad (22)$$




Step Force with finite Rise time

$$P(t) = P_0\frac{t}{t_r}, \quad t \le t_r$$

$$P(t) = P_0, \quad t \ge t_r$$




Phase-I: Response during Ramp

$$u(t) = \frac{P_0}{K}\left[\frac{t}{t_r} - \frac{\sin\omega t}{\omega t_r}\right], \quad t \le t_r \quad (23)$$





Phase-II: Response during constant Phase (+)

From Duhamel integral,

$$u(t) = \frac{1}{m\omega}\int_0^t P(\tau)\sin\omega(t-\tau)\,d\tau$$

$$= \frac{P_0}{K}\left[1-\cos\omega t\right]$$

$$\therefore u(t-t_r) = \frac{P_0}{K}\left[1-\cos\omega(t-t_r)\right], \quad t-t_r \ge 0 \quad (24)$$

At the beginning of second phase, the initial conditions are not zero.

i.e. at $t=t_r$, $u(t_r) \neq 0$ & $\dot{u}(t_r) \neq 0$

$$\therefore u(t) = u(t_r)\cos\omega(t-t_r) + \frac{\dot{u}(t_r)}{\omega}\sin\omega(t-t_r) + \frac{P_0}{K}\left[1-\cos\omega(t-t_r)\right] \quad (25)$$

[Free vibration eqn: $u = A\cos\omega t + B\sin\omega t$] $t \ge t_r$

Free vibration sol$^{n}$: $u(t) = A\cos\omega t + B\sin\omega t$ $\quad t \geq t_r$

$A = u(0)$ & $B = \frac{\dot{u}(0)}{\omega}$

But $u(0) \neq 0$ have, $u(0) = u(t_r)$

& $\dot{u}(0) \neq 0$ but $\dot{u}(0) = \dot{u}(t_r)$

$\therefore u(t) = u(t_r)\cos\omega(t-t_r) + \frac{\dot{u}(t_r)}{\omega}\sin\omega(t-t_r) + \frac{P_0}{K}[1-\cos\omega(t-t_r)]$ — (25)

from eq$^{n}$ (23)

$\dot{u}(t_r) = \frac{P_0}{K}\left[\frac{1}{t_r} - \frac{\omega\cos\omega t_r}{\omega t_r}\right] = \frac{P_0}{Kt_r}(1-\cos\omega t_r)$

$\therefore u(t_r) = \frac{P_0}{K}\left(\frac{t_r}{t_r} - \frac{\sin\omega t_r}{\omega t_r}\right) = \frac{P_0}{K}\left(1 - \frac{\sin\omega t_r}{\omega t_r}\right)$

& $\dot{u}(t_r) = \frac{P_0}{Kt_r}(1-\cos\omega t_r)$

Substituting & solving eq$^{n}$ (25)

$u(t) = \frac{P_0}{K}\left[1 + \frac{1}{\omega t_r}\left[(1-\cos\omega t_r)\sin\omega(t-t_r) - \sin\omega t_r\cos\omega(t-t_r)\right]\right]$ — (26), $t \geq t_r$

or $u(t) = \frac{P_0}{K}\left[1 - \frac{1}{\omega t_r}\left[\sin\omega t - \sin\omega(t-t_r)\right]\right]$ — (27), $t \geq t_r$



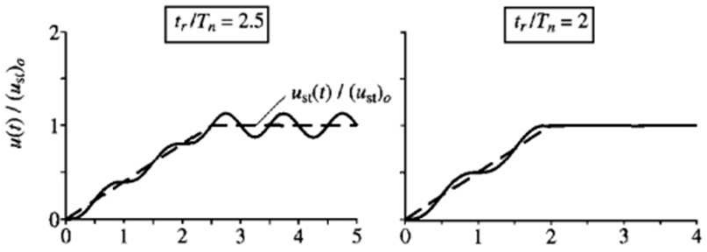


The normalized deformation $\frac{u(t)}{u_{st}}$ is a function of the normalized time, $\frac{t}{T}$ .

Because, $\omega t = \frac{2\pi t}{T} = 2\pi\left(\frac{t}{T}\right)$

This function depends only on the ratio $\frac{t_r}{T}$ , because $\omega t_r = 2\pi\left(\frac{t_r}{T}\right)$

not separately on tr and T.



## 2.8 Convolution Integral and Duhamel Integral

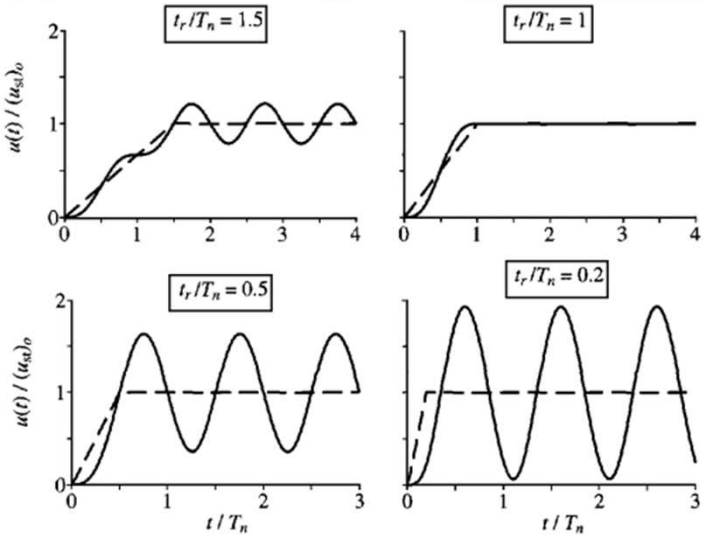



**Figure 4.5.2** Dynamic response of undamped SDF system to step force with finite rise time; static solution is shown by dashed lines.



## 2.8 Convolution Integral and Duhamel Integral

1. During the force-rise phase the system oscillates at the natural period $T_n$ about the static solution.
2. During the constant-force phase the system oscillates also at the natural period $T_n$ about the static solution.
3. If the velocity $\dot{u}(t_r)$ is zero at the end of the ramp, the system does not vibrate during the constant-force phase.
4. For smaller values of $t_r/T_n$ (i.e., short rise time), the response is similar to that due to a sudden step force; see Fig. 4.3.1c.
5. For larger values of $t_r/T_n$, the dynamic displacement oscillates close to the static solution, implying that the dynamic effects are small (i.e., a force increasing slowly—relative to $T_n$—from 0 to $p_o$ affects the system like a static force).

max value of u(t) during constant force

$$u_{max} = \frac{p_0}{k}\left[1 + \frac{1}{\omega t_r}\sqrt{(1-\cos\omega t_r)^2 + \sin^2\omega t_r}\right]$$

and, Response factor, $R_d = \frac{u_{max}}{u_{st}} = 1 + \frac{\left|\sin\frac{\pi t_r}{T}\right|}{\frac{\pi t_r}{T}}$

## 2.8 Convolution Integral and Duhamel Integral

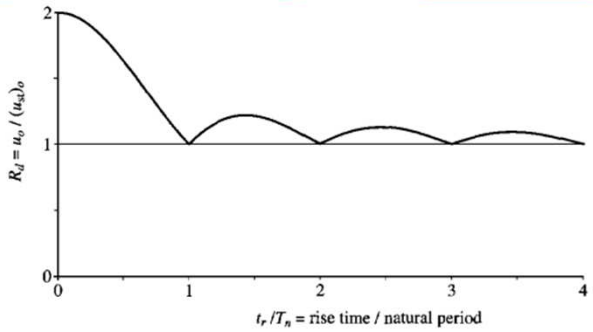



**Figure 4.5.3** Response spectrum for step force with finite rise time.

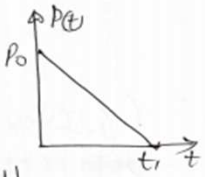
1. If $t_r < T_n/4$ (i.e., a relatively short rise time), $u_o \simeq 2(u_{st})_o$, implying that the structure "sees" this excitation like a suddenly applied force.
2. If $t_r > 3T_n$ (i.e., a relatively long rise time), $u_o \simeq (u_{st})_o$, implying that this excitation affects the structure like a static force.
3. If $t_r/T_n = 1, 2, 3, \ldots, \dot{u}(t_r) = 0$ at the end of the force-rise phase, and the system does not oscillate during the constant-force phase; therefore, $u_o = (u_{st})_o$.



## 2.8 Convolution Integral and Duhamel Integral

#① For a SDOF system is subjected to a triangular pulse load of amplitude $P_0$ the during $t_1$ as shown in fig. below. Determine the response of the undamped system during forced vibration and free vibration phase. ①

Soln: $P(\tau) = P_0\left(1 - \frac{\tau}{t_1}\right)$ for $0 \le \tau \le t_1$




Duhamel's integral:- (Response to arbitrary load)

$u(t) = \frac{1}{m\omega}\int P(\tau)\sin\omega(t-\tau)d\tau$, undamped, start from rest

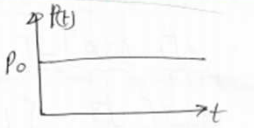
$u(t) = \frac{1}{m\omega}\int P(\tau)\sin\omega(t-\tau)d\tau + u_0\cos\omega t + \frac{\dot{u}_0}{\omega}\sin\omega t$, undamped

## 2.8 Convolution Integral and Duhamel Integral

Undamped:

Constant force!

$P(t) = P_0$

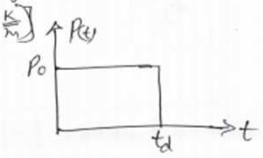



$$\therefore v(t) = \frac{1}{m\omega}\int_0^t P(\tau)\sin\omega(t-\tau)\,d\tau$$

$$= \frac{P_0}{m\omega}\int_0^t \sin\omega(t-\tau)\,d\tau = \frac{P_0}{m\omega^2}\left[\cos\omega(t-\tau)\right]_0^t = \frac{P_0}{m\omega^2}(1-\cos\omega t)$$

$$= \frac{P_0}{K}(1-\cos\omega t) \quad \left[\omega^2 = \frac{K}{m}\right]$$

Rectangular force




$P(t) = P(0) \quad 0 < \tau \le t_d$

$P(t) = 0 \quad \tau > t_d$

upto $\tau \le t_d$.

$$v(t) = \frac{1}{m\omega}\int_0^{t_d} P(\tau)\sin\omega(t-\tau)\,d\tau = \frac{P_0}{m\omega^2}(1-\cos\omega t_d) = \frac{P_0}{K}(1-\cos\omega t_d)$$

$$\& \quad \dot{v}(t_d) = \frac{P_0}{K}\omega\sin\omega t_d$$

For the response after time $t_d$ we apply free vibration, taking as the initial conditions the displacement and velocity at $t_d$.

$$v(0) = v(t_d) = \frac{P_0}{K}(1-\cos\omega t_d) \quad \& \quad \dot{v}(0) = \dot{v}(t_d) = \frac{P_0}{K}\omega\sin\omega t_d$$



## 2.8 Convolution Integral and Duhamel Integral

$$v(t) = \frac{1}{m\omega}\int P(\tau)\sin\omega(t-\tau)\,d\tau + v(0)\cos\omega t + \frac{\dot{v}(0)}{\omega}\sin\omega t, \text{ undamped.}$$

$$v(t) = v(0)\cos\omega(t-t_d) + \frac{\dot{v}(0)}{\omega}\sin\omega(t-t_d)$$

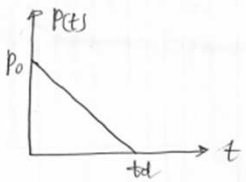
$$= \frac{P_0}{K}(1-\cos\omega t_d)\cos\omega(t-t_d) + \frac{1}{\omega}\times\frac{P_0}{K}\omega\sin\omega t_d\cdot\sin\omega(t-t_d)$$

$$= \frac{P_0}{K}(1-\cos\omega t_d)\cos\omega(t-t_d) + \frac{P_0}{K}\sin\omega t_d\cdot\sin\omega(t-t_d)$$

Can be reduced to

$$v(t) = \frac{P_0}{K}\{\cos\omega(t-t_d) - \cos\omega t\}$$

Triangular Load!




$P(\tau) = P_0\left(1-\frac{\tau}{t_d}\right)$ for $\tau \le t_d$

for $\tau \le t_d$.

$$v(t) = \frac{1}{m\omega}\int_0^t P(\tau)\sin\omega(t-\tau)\,d\tau = \frac{P_0}{m\omega}\int_0^t\left(1-\frac{\tau}{t_d}\right)\sin\omega(t-\tau)\,d\tau$$

## 2.8 Convolution Integral and Duhamel Integral

$= \frac{p_0}{m\omega^2}(1-\cos\omega t) + \frac{p_0}{m\omega^2 t_d}\cdot\left(\frac{\sin\omega t}{\omega} - t\right)$

$= \frac{p_0}{k}(1-\cos\omega t) + \frac{p_0}{k t_d}\left(\frac{\sin\omega t}{\omega} - t\right)$ for $t \le t_d$

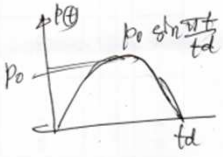
and, $\dot{v}(t_d) = \frac{p_0}{k}\cdot\omega\sin\omega t + \frac{p_0}{k t_d}\cdot\cos\omega t - 1$

For response after time $t_d$, we apply free vibration

$v(t) = v(t_d)\cdot\cos\omega(t-t_d) + \frac{\dot{v}(t_d)}{\omega}\sin\omega(t-t_d)$

Half Sine wave load

$p(t) = p_0 \sin\frac{\pi t}{t_d}$




for $t \le t_d$

$v(t) = \frac{1}{m\omega}\int_0^t p(\tau)\sin\omega(t-\tau)\,d\tau = \frac{1}{m\omega}\int_0^t p_0 \sin\frac{\pi\tau}{t_d}\cdot\sin\omega(t-\tau)\,d\tau$

$= \frac{p_0}{m\omega}\int_0^t \sin\frac{\pi\tau}{t_d}\cdot\sin\omega(t-\tau)\,d\tau$



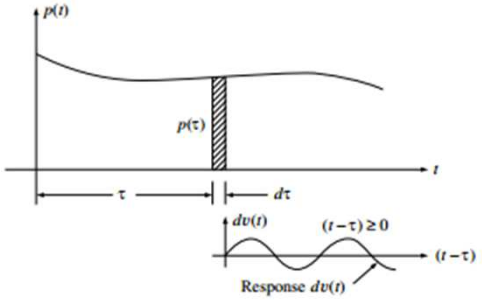
## 2.8 Forced Vibration Response to General Dynamic Loading

Response to General Dynamic Loading: Superposition Method

Undamped System:

Short duration impulsive loads can be used as a basis of developing a formula for evaluating response to a general dynamic loading.

Consider an arbitrary general loading p(t) as shown. At time t = τ, intensity of loading would be p(τ) acting at time t = τ.



## 2.8 Forced Vibration Response to General Dynamic Loading

This loading acting during the interval of time dτ represents a very short duration impulse

$p(\tau)d\tau$, on the structure.

So that Duhamel integral can be used to evaluate the response.

As this method is approximate for impulses of finite duration, it becomes exact as the duration of loading approaches zero.

Thus for the differential time interval dτ, the response produced by the impulse $p(\tau)d\tau$ is exactly equal to

$$dv(t) = \frac{p(\tau)\,d\tau}{m\,\omega}\,\sin\omega\,(t-\tau) \qquad t \geq \tau$$



## 2.8 Forced Vibration Response to General Dynamic Loading

In this expression, the term du(t) represents the time-history response to the differential impulse over the time $t \geq \tau$; it is not the change of u during a time interval dt.

The entire loading history can be considered to consist of a succession of such short impulses.

For the linearly elastic system, the total response can be obtained by summing all the differential responses.

$$v(t) = \frac{1}{m\,\omega}\int_0^t p(\tau)\,\sin\omega\,(t-\tau)\,d\tau \qquad t \geq 0$$

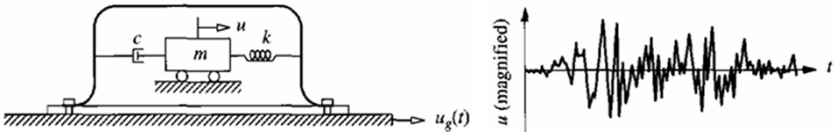
Similarly for Damped system

$$v(t) = \frac{1}{m\,\omega_D}\int_0^t p(\tau)\,\sin\omega_D\,(t-\tau)\,\exp\left[-\xi\,\omega\,(t-\tau)\right]d\tau \qquad t \geq 0$$

## Vibration Measuring Instruments

A simplest form of a transducer is a mass-spring-damper system mounted inside a rigid frame that is attached to the ground surface.

When subjected to motion of the support, the transducer mass moves relative to frame, and this relative displacement is recorded after suitable magnification.




$M\ddot{u}(t)+c\dot{u}(t)+ku(t) = P_{eff}(t) = -m\ddot{u}_g(t)$

$= -m\ddot{u}_{go}Sin\varpi t \quad \ldots. \quad (6)$





Steady state solution,

$$u(t) = -\frac{m\ddot{u}_{go}}{k} R_d \cdot \sin(\bar{\omega}t-\theta) \quad \text{---} \quad (4)$$

Alternatively,

$u_g(t) = u_{go} \cdot \sin\bar{\omega}t$

$\therefore \dot{u}_g(t) = \bar{\omega} u_{go} \cdot \cos\bar{\omega}t$

& $\ddot{u}_g(t) = -\bar{\omega}^2 u_{go} \cdot \sin\bar{\omega}t$

$\therefore P_{eff} = -m \cdot -\bar{\omega}^2 u_{go} \cdot \sin\bar{\omega}t$

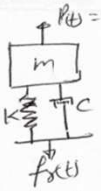
$= +m\bar{\omega}^2 u_{go} \cdot \sin\bar{\omega}t$

$\therefore u(t) = \frac{m\bar{\omega}^2 u_{go}}{k} R_d \sin(\bar{\omega}t-\theta) = \frac{\bar{\omega}^2}{\omega^2} u_{go} \cdot R_d \sin(\bar{\omega}t-\theta)$

$$\boxed{u(t) = \beta R_d u_{go} \sin(\bar{\omega}t-\theta)} \quad \text{---} \quad (7)$$

## 2.10 Vibration isolation (Vertical oscillating force, harmonic motion of the base, force transmissibility, unbalanced mass amplitude of rotating unbalanced system)

Consider mass-spring-damper system. (III)




$f_S(t) = k\,u(t)$

$f_D(t) = c\,\dot{u}(t)$

& $f_T(t) = f_S(t) + f_D(t) = k\,u(t) + c\,\dot{u}(t)$

But, $u(t) = \frac{P_0}{K} Rd \cdot \sin(\bar{\omega}t - \theta)$ & $\dot{u}(t) = \frac{P_0}{K} Rd \cdot \bar{\omega}\cos(\bar{\omega}t - \theta)$

$\therefore f_T(t) = P_0 Rd \sin(\bar{\omega}t - \theta) + \frac{P_0}{K} Rd \cdot c\bar{\omega}\cos(\bar{\omega}t - \theta)$ — (H-1)

The maximum value of $f_T(t)$ over time 't' is

$$f_{T(max)} = \left[f_{S(max)}^2 + f_{D(max)}^2\right]^{1/2}$$

$$= \sqrt{(P_0 Rd)^2 + \left(\frac{P_0}{K} \cdot Rd\, c\bar{\omega}\right)^2}$$



## 2.10 Vibration isolation

$$= P_0 Rd \sqrt{1 + \left(\frac{2m\bar{\omega}\bar{\omega}\xi}{K}\right)^2} \quad [\because c = 2m\omega\xi]$$

$$= P_0 Rd \sqrt{1 + \left(\frac{2\,\omega\bar{\omega}\xi}{\omega^2}\right)^2} \quad \left[\frac{K}{m} = \omega^2\right]$$

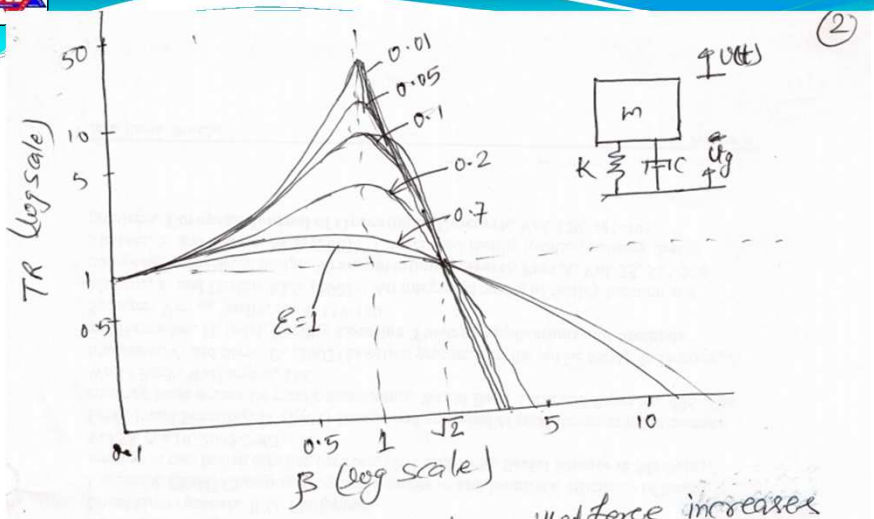
$$f_{T(max)} = P_0 Rd \sqrt{1 + (2\xi\beta)^2} \quad \text{(H2)} \quad \left[\because \beta = \frac{\bar{\omega}}{\omega}\right]$$

Transmissibility of force = $\frac{\text{Max. transmitted Force}}{\text{Amplitude of Force}}$

or $TR = \frac{f_{T(max)}}{P_0} = Rd\sqrt{1 + (2\xi\beta)^2}$

or $TR = \frac{\sqrt{1 + (2\xi\beta)^2}}{\sqrt{(1-\beta^2)^2 + (2\xi\beta)^2}} = \left[\frac{1 + (2\xi\beta)^2}{(1-\beta^2)^2 + (2\xi\beta)^2}\right]^{1/2}$ — (H3)

## 2.10 Vibration Isolation




(2)

Damping decreases → transmitted force increases

only if $\omega/\omega < \sqrt{2}$

transmitted force to be less than the



## 2.10 Vibration Isolation

only if $\omega/\omega < \sqrt{2}$

For the transmitted force to be less than the applied force, the stiffness of the support system and hence the natural frequency should be small enough, so that $\omega/\omega > \sqrt{2}$.

Natural rubber is a very satisfactory material for base isolation.

Harmonic ground motion,

$$\ddot{u}_g(t) = \ddot{u}_{g0} \cdot \sin\omega t$$

$$\& \quad u(t) = \frac{-m\ddot{u}_{g0}}{K} R_d \sin(\omega t - \theta) \quad \text{——(4)}$$

$$\& \quad TR = R_d\sqrt{1+(2\xi\beta)^2} = \quad \text{——(5)}$$

## 2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

Computer simulation is the process of using a computer program to imitate the behavior of a real-world system or process. It helps people study, predict, or test situations without performing them in real life.

For example:
Weather forecasting uses simulations to predict rain or storms.
Flight simulators train pilots safely.
Scientists simulate the movement of planets, diseases, or traffic systems.
Engineers test car designs through computer simulations before building actual cars.
In simple terms, a computer simulation creates a virtual model of something real so we can observe what might happen under different conditions.



## 2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

In simple terms, a computer simulation creates a virtual model of something real so we can observe what might happen under different conditions.

**Main features of computer simulation**
**Modeling** – Creating a mathematical or logical model of a system.
**Experimentation** – Changing inputs or conditions to see outcomes.
**Prediction** – Estimating future behavior or performance.
**Cost and safety saving** – Avoiding expensive or dangerous real-life testing.

## 2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

Computer simulation of the dynamic response of a **Single-Degree-of-Freedom (SDOF)** system involves using numerical methods to calculate how the system responds to dynamic loads such as earthquakes, wind, impact, or harmonic forces.

The equation of motion is:

$$m\ddot{u}(t)+c\dot{u}(t)+ku(t) = P(t) \quad \ldots \quad \ldots \quad (1)$$

<u>The simulation is used to determine:</u>

Displacement response, Velocity response, Acceleration response, Maximum response values, Effect of damping and loading characteristics



## 2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

**Steps in Computer Simulation**

Define system properties:

- Mass, Stiffness, Damping coefficient

Define loading function:

- Harmonic load, Step load, Pulse load, Earthquake ground motion

Choose a numerical integration method:

- Newmark-β Method
- Central Difference Method
- Runge-Kutta Method
- Wilson-θ Method

Divide time into small intervals Δt

Compute displacement, velocity, and acceleration at each time step.

Plot response-time histories and analyze results.

## 2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

A computer program (e.g., MATLAB, Python, OpenSees, SAP2000) calculates the response at each time step and generates displacement, velocity, and acceleration histories.

**Example: Computer Simulation of an SDOF System**
Consider an SDOF system with:
Mass, m = 1000 kg
Stiffness, k = 40,000 N/m
Damping ratio, ζ = 0.05 (5%)
External force:
F(t)=1000sin(5t) N



2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

**Step 1: Calculate Natural Frequency**

$$w = \sqrt{k/m} = \sqrt{40000/1000} = \sqrt{40} = 6.32\ rad/s$$

**Step 2: Calculate Damping Coefficient**
c=2ξmω = 2(0.05)(1000)(6.32) = 632 N\c

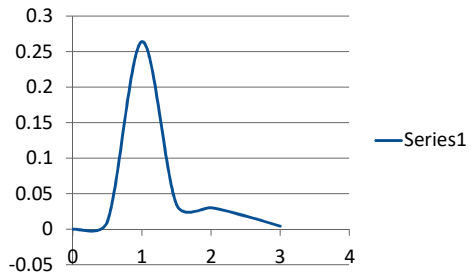
**Step 3: Equation of Motion**
1000ü(t)+632ů(t)+40000u(t) = 1000sin(5t)

**Step 4: Computer Simulation**
Using a numerical method (e.g., Newmark-β or Runge-Kutta), the response is calculated at small time intervals.

2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect

**A typical output may be:**

| Time | Displacement |
|---|---|
| 0.0 | 0.000 |
| 0.5 | 0.0108 |
| 1.0 | 0.0264 |
| 1.5 | 0.0347 |
| 2.0 | 0.0302 |
| 2.5 | 0.0185 |
| 3.0 | 0.0043 |






आओ गीता पढ़े!
निःशुल्क (Online)
**19th June 2026** से 48 वें बैच का शुभारंभ!

🎉 आज ही पंजीयन करें। 🎉
⬇️⬇️⬇️⬇️⬇️
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।। गीता पढ़ें, पढ़ायें, जीवन में लायें ।। 13:24




# THANK YOU

## 2.15 Time Domain Analysis

A time domain analysis is an analysis of physical signals, mathematical functions, or time series of economic or environmental data, in reference to time.

In the time domain, function's value is understood for all real numbers at various separate instances in the case of discrete-time or the case of continuous-time.

A time-domain graph can show how a signal changes with time.

In general, when an analysis uses a unit of time, such as seconds or one of its multiples (minutes or hours) as a unit of measurement, then it is in the time domain.

2.11 Computer simulation of dynamic responses of single-degree-of-freedom system: Free, Forced, Undamped, Damped (Under critical, critical and over critical); Response effect



## 2.16 Frequency Domain Analysis

Frequency domain is an analysis of signals or mathematical functions, in reference to frequency, instead of time.

Frequency domain graph displays the how much of the signal exists within a given frequency band concerning a range of frequencies.

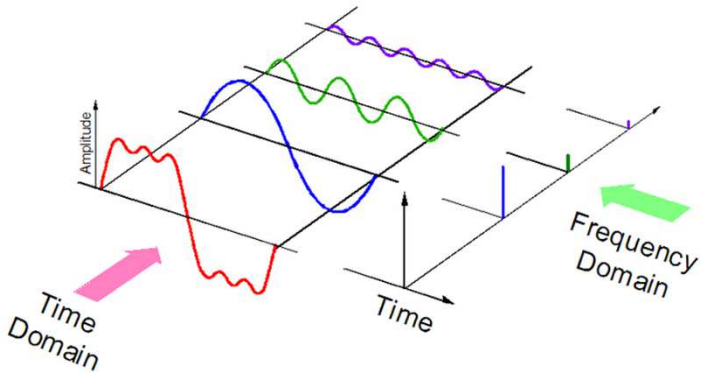
A frequency-domain representation can include information on the phase shift that must be applied to each sinusoid to be able to recombine the frequency components to recover the original time signal.

Frequency-domain analysis concerns the units like Hertz, then it is in the frequency domain.

## 2.16 Frequency Domain Analysis

The Fourier transform relates the function in the time domain, to the function in the frequency domain.

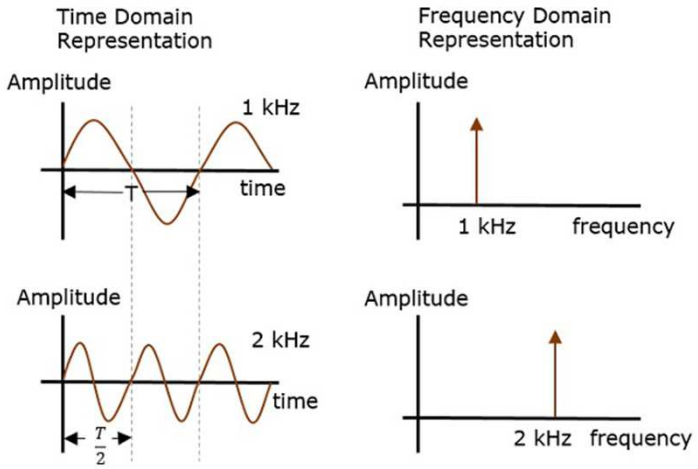
The component frequencies, spread across the frequency spectrum, are represented as peaks in the frequency domain.




https://www.kaggle.com/gabbygab/audio-visualization-case-study/notebook



## 2.16 Frequency Domain Analysis



## Spectrum Examples

